

Highlighted Manuscript: Change and Deletion Index

Baseline: the manuscript as submitted through the IEEE Access portal. Yellow marks show inserted/replaced rendered words; orange frames identify pages containing changed figure assets. The clean PDF text is unchanged by annotations. Source-only layout/preamble edits are preserved in reviewed-baseline-source.diff. This index records removed text that cannot be highlighted in the revised manuscript.

Change 1: replace | old pp. [1] | new pp. [1]

Before (including deletions): SPICE

After: Structural Circuit

Change 2: replace | old pp. [1] | new pp. [1]

Before (including deletions): Automated digitization of

After: Recovering a netlist from a

Change 3: replace | old pp. [1] | new pp. [1]

Before (including deletions): circuit schematics into SPICE netlists remains challenging because connectivity recovery is brittle under noisy

After: schematic requires deciding which

Change 4: replace | old pp. [1] | new pp. [1]

Before (including deletions): detections.

After: strokes connect which component terminals.

Change 5: replace | old pp. [1] | new pp. [1]

Before (including deletions): that converts a scanned schematic into a SPICE netlist through component detection, an

After: for this structural connectivity task. An

Change 6: insert | old pp. [] | new pp. [1]

Before (including deletions): (none)

After: extractor masks components before binarization and returns

Change 7: replace | old pp. [1] | new pp. [1]

Before (including deletions): extractor,

After: segments with explicit endpoints. A typed endpoint graph links endpoints, pins and wire bodies under scale-relative tolerances,

Change 8: replace | old pp. [1] | new pp. [1]

Before (including deletions): graph-based

After: degree-budget completion step reconnects floating pins by bounded assignment. We evaluate on two real- image connectivity benchmarks from CGHD-1152: 31 images with human-verified nets, and 164 held-out photographs from 24 of the corpus's 25 drafters whose reference nets we derive from the dataset's own stroke and symbol annotations. With annotated component boxes, the

Change 9: insert | old pp. [] | new pp. [1]

Before (including deletions): (none)

After: reaches component-pair micro-F1 0.890 on the human-verified set, on which its configuration was selected, and 0.711 on the held-out photographs. On both benchmarks it outperforms every deterministic baseline, including connected-component tracing and Hough line fitting, in paired tests with Holm correction. With annotated wires, its score stays within 0.005 of the native value when the images are rescaled from 0.5× to 3×. Replacing the annotated boxes with a trained component detector gives micro-F1 0.627, and most of

Change 10: replace | old pp. [1] | new pp. [1]

Before (including deletions): resolves electrical connectivity.

After: loss comes from missed components.

Change 11: replace | old pp. [1] | new pp. [1]

Before (including deletions): main contribution

After: wire extractor reaches F1 0.973 on 134 annotated images; on the held-out photographs, wire extraction

Change 12: replace | old pp. [1] | new pp. [1]

Before (including deletions): joining stage: an endpoint-graph

After: main source of error. A vision-language

Change 13: replace | old pp. [1] | new pp. [1]

Before (including deletions): over wire endpoints and component pins with five edge types, combined with degree-budget completion that recovers dropped connections through min-cost b-matching while bounding over-merge with per-pin budgets and self-loop guards. We also introduce a synthetic perturbation protocol and, to our knowledge,

After: given

Change 14: replace | old pp. [1] | new pp. [1]

Before (including deletions): first human-verified net-level connectivity benchmark for hand-drawn circuits, covering 31 images from CGHD-1152. The occlusion-first extractor

After: same electrical-component boxes

Change 15: replace | old pp. [1] | new pp. [1]

Before (including deletions): wire-detection F1 = 0.976 on 134 scanned images. On the human-verified net-level benchmark, the best join reaches connectivity micro- F1 = 0.890, outperforming the prior completion default (0.83), a legacy radius union-find join (0.67), a connected-component net-tracing baseline (0.62), and a Hough-plus-proximity line tracer (0.81), all under identical detected wires. Running the join on perfect wire labels does not improve micro-F1, showing that connectivity, not wire detection, remains the bottleneck. A first-party comparison shows a strong vision- language model reaches

After: micro-F1

Change 16: replace | old pp. [1] | new pp. [1]

Before (including deletions): same images, statistically indistinguishable

After: human-verified set: the pooled difference

Change 17: replace | old pp. [1] | new pp. [1]

Before (including deletions): but far more expensive

After: is not significant,

Change 18: replace | old pp. [1] | new pp. [1]

Before (including deletions): without structural guarantees, supporting explicit geometric connectivity recovery for deployable circuit digitization.

After: per- image scores favor the model. We release the code, both benchmarks and the evaluation scripts.

Change 19: replace | old pp. [1] | new pp. [1]

Before (including deletions): SPICE,

After: structural netlist,

Change 20: replace | old pp. [1] | new pp. [1]

Before (including deletions): Analog

After: A

Change 21: replace | old pp. [1] | new pp. [1]

Before (including deletions): schematics encode functional information (component

After: schematic encodes component

Change 22: replace | old pp. [1] | new pp. [1]

Before (including deletions): values,

After: component values

Change 23: replace | old pp. [1] | new pp. [1]

Before (including deletions): topological connections)

After: the connections between components

Change 24: replace | old pp. [1] | new pp. [1]

Before (including deletions): visual format designed

After: drawing made

Change 25: replace | old pp. [1] | new pp. [1]

Before (including deletions): human interpretation. Converting scanned

After: people. A netlist records the same connections in a form that software can check, simulate and search. Converting hand-drawn

Change 26: delete | old pp. [1] | new pp. []

Before (including deletions): machine-readable SPICE (Simulation Program with Integrated Circuit Emphasis)

After: (none)

Change 27: replace | old pp. [1] | new pp. [1]

Before (including deletions): enables automated simulation, design verification,

After: would let course material, lab notebooks

Change 28: replace | old pp. [1] | new pp. [1]

Before (including deletions): large-scale circuit analysis. While recent advances

After: legacy designs enter these tools without manual re-entry. This paper addresses the structural part of that conversion: given the components

Change 29: replace | old pp. [1] | new pp. [1]

Before (including deletions): object detection have made component recognition increasingly reliable, the joining problem, determining which wire segments connect to

After: an image, decide

Change 30: replace | old pp. [1] | new pp. [1]

Before (including deletions): pins and to each other, remains the primary failure mode in end-to-end digitization pipelines. The challenge

After: terminals share an electrical node. We call this step joining. Joining

Change 31: replace | old pp. [1] | new pp. [1]

Before (including deletions): threefold.

After: hard for three reasons.

Change 32: delete | old pp. [1] | new pp. []

Before (including deletions): inherently

After: (none)

Change 33: replace | old pp. [1] | new pp. [1]

Before (including deletions): produces endpoint positions

After: yields endpoints at different off-sets from the pins, and strokes stop short of or overshoot their targets. Second, component symbols overlap the wire endpoints

Change 34: replace | old pp. [1] | new pp. [1]

Before (including deletions): vary by 10–80 pixels. Second, components occupy bounding boxes that overlap with wire endpoints, creating ambiguity about whether

After: attach to them, so

Change 35: replace | old pp. [1] | new pp. [1]

Before (including deletions): connects

After: near a component may belong

Change 36: replace | old pp. [1] | new pp. [1]

Before (including deletions): a component's pin

After: one of its pins

Change 37: replace | old pp. [1] | new pp. [1]

Before (including deletions): passes nearby.

After: pass by.

Change 38: replace | old pp. [1] | new pp. [1]

Before (including deletions): fragments,

After: fragments

Change 39: replace | old pp. [1] | new pp. [1]

Before (including deletions): crossing wires create complex connectivity patterns that simple radius-based approaches cannot resolve without over-merging.

After: wire crossings all look alike to a proximity rule.

Change 40: replace | old pp. [2] | new pp. [2]

Before (including deletions): YOLO26m

After: YOLO

Change 41: replace | old pp. [2] | new pp. [2]

Before (including deletions): SPICE

After: Structural

Change 42: replace | old pp. [2] | new pp. [2]

Before (including deletions): Pin discovery + simulation

After: Inferred pin-to-node map

Change 43: replace | old pp. [2] | new pp. [2]

Before (including deletions): End-to-end pipeline

After: Pipeline

Change 44: replace | old pp. [2] | new pp. [2]

Before (including deletions): scanning circuit diagram digitization. Our contributions span the

After: structural circuit-netlist recovery. The highlighted stages extract

Change 45: replace | old pp. [2] | new pp. [2]

Before (including deletions): extraction

After: endpoints

Change 46: delete | old pp. [2] | new pp. []

Before (including deletions): joining stages; the

After: (none)

Change 47: replace | old pp. [2] | new pp. [2]

Before (including deletions): (dashed red) determines which extracted wire segments connect to which component pins and is the dominant failure mode in end-to-end digitization.

After: them into nets.

Change 48: delete | old pp. [2] | new pp. []

Before (including deletions): real

After: (none)

Change 49: replace | old pp. [2] | new pp. [2]

Before (including deletions): circuit images,

After: circuits,

Change 50: replace | old pp. [2] | new pp. [2]

Before (including deletions): our

After: the

Change 51: replace | old pp. [2] | new pp. [2]

Before (including deletions): (connectivity

After: (component-pair

Change 52: replace | old pp. [2] | new pp. [2]

Before (including deletions): (36

After: (35

Change 53: replace | old pp. [2] | new pp. [2]

Before (including deletions): 12 nets); right:

After: 7 nets). Right:

Change 54: replace | old pp. [2] | new pp. [2]

Before (including deletions): (12

After: (11

Change 55: replace | old pp. [2] | new pp. [2]

Before (including deletions): shows, in order: original scan,

After: shows the input,

Change 56: replace | old pp. [2] | new pp. [2]

Before (including deletions): close,

After: closing,

Change 57: replace | old pp. [2] | new pp. [2]

Before (including deletions): (red overlay),

After: (red)

Change 58: replace | old pp. [2] | new pp. [2]

Before (including deletions): every recovered

After: each

Change 59: replace | old pp. [2] | new pp. [2]

Before (including deletions): is a distinct colour drawn along

After: has

Change 60: replace | old pp. [2] | new pp. [2]

Before (including deletions): conductors; component bounding

After: own color;

Change 61: replace | old pp. [2] | new pp. [2]

Before (including deletions): ground-truth labels (as used throughout the net-level evaluation, isolating the join),

After: annotated components,

Change 62: replace | old pp. [2] | new pp. [2]

Before (including deletions): mark junctions,

After: are junctions

Change 63: replace | old pp. [2] | new pp. [2]

Before (including deletions): mark

After: are

Change 64: insert | old pp. [] | new pp. [2]

Before (including deletions): (none)

After: Most image-to-netlist systems erase the components and read nets from the connected components or contours of the remaining wire pixels [1], [2].

Change 65: replace | old pp. [2] | new pp. [2]

Before (including deletions): natural alternative is

After: single stroke gap then splits a net, and a single touching stroke merges two. Other systems pass the image

Change 66: delete | old pp. [2] | new pp. []

Before (including deletions): prompt

After: (none)

Change 67: replace | old pp. [2] | new pp. [2]

Before (including deletions): vision-language

After: language

Change 68: replace | old pp. [2] | new pp. [2]

Before (including deletions): (VLM) to

After: (LLM) and

Change 69: replace | old pp. [2] | new pp. [2]

Before (including deletions): the schematic

After: connectivity from its answer [3]. We keep an explicit representation of every wire endpoint

Change 70: replace | old pp. [2] | new pp. [2]

Before (including deletions): emit its netlist directly. We test this in Section IV-F: a strong VLM attains connectivity accuracy statistically indistinguishable from our

After: decide each connection with

Change 71: replace | old pp. [2] | new pp. [2]

Before (including deletions): pipeline, but it does so at orders-of-magnitude higher cost, returns free-form text rather than a structured netlist, offers no structural guarantee against electrically invalid output (e.g. shorted two-terminal parts), and drops connections on the most complex circuits. These properties motivate an explicit, inspectable geometric pipeline whose output is a netlist by construction and whose accuracy matches the VLM's while being deterministic and structurally valid by construction. We address these challenges with a complete deterministic pipeline whose novel elements are: (1) an

After: rules that can be inspected. This paper makes four contributions: 1) An

Change 72: replace | old pp. [2] | new pp. [2]

Before (including deletions): masks detected components

After: fills each component region with its local median color

Change 73: replace | old pp. [2] | new pp. [2]

Before (including deletions): binarization

After: Sauvola binarization

Change 74: replace | old pp. [2] | new pp. [2]

Before (including deletions): estimates per-segment endpoints, producing a line-segment-with-endpoints representation (rather than tracing nets directly into connected components, as prior systems do) that is what makes graph-based joining possible; (2) an

After: returns line segments whose endpoints are estimated by principal component analysis (PCA). It reaches wire-detection F1 0.973 on 134 CGHD-1152 images (0.976 for the best of 36 variants). 2) A typed

Change 75: delete | old pp. [2] | new pp. []

Before (including deletions): model that constructs a graph over both wire endpoints and component pins,

After: (none)

Change 76: replace | old pp. [2] | new pp. [2]

Before (including deletions): types capturing distinct

After: rules, component-first pin assignment and scale-relative tolerances, followed by degree-budget completion, which reconnects floating pins through a bounded-capacity assignment with a shared-component guard. 3) Two real-image

Change 77: replace | old pp. [2] | new pp. [2]

Before (including deletions): patterns; and (3) a degree-budget completion algorithm that identifies and reconnects dropped connections using constrained optimization. We evaluate on the CGHD-1152 dataset [1]

After: benchmarks: 31 CGHD- 1152 images

Change 78: replace | old pp. [2] | new pp. [2]

Before (including deletions): both a synthetic error injection framework that enables systematic robustness analysis

After: human-verified nets,

Change 79: replace | old pp. [2] | new pp. [2]

Before (including deletions): human-verified net-level benchmark.

After: 164- image held-out benchmark covering 24 of the 25 CGHD drafters whose reference nets are derived from the CGHD authors' own stroke-segmentation maps and symbol polygons. The derived reference agrees with the human nets on 17 overlapping images at micro-F1 0.939, and a human audit of 22 of 40 randomly drawn held-out images, mostly the smaller circuits, found it exact on 21. 4) An evaluation protocol with paired significance tests under Holm correction, complexity and component- size strata, controlled image rescaling, end-to-end runs with a trained component detector, and a vision- language model (VLM) reference that receives the electrical-component boxes given to our join. The output is a structural netlist; recognizing component values and device models is outside the scope of this work. Section II reviews related work, Section III describes the method and Section IV the experimental setup. Section V reports results, and Section VI discusses them together with the scope and limitations of the study.

Change 80: replace | old pp. [2] | new pp. [2]

Before (including deletions): The end-to-end conversion of

After: Several systems convert

Change 81: replace | old pp. [2] | new pp. [2]

Before (including deletions): machine-readable netlists has progressed rapidly.

After: machine- readable netlists.

Change 82: replace | old pp. [2] | new pp. [2]

Before (including deletions): [3]

After: [4]

Change 83: replace | old pp. [2] | new pp. [2]

Before (including deletions): pattern-recognition

After: pattern- recognition

Change 84: replace | old pp. [2] | new pp. [2]

Before (including deletions): digitizing circuit

After: printed

Change 85: replace | old pp. [2] | new pp. [2]

Before (including deletions): combining

After: combining

Change 86: replace | old pp. [2] | new pp. [2]

Before (including deletions): [4], the closest contemporary system,

After: [1]

Change 87: replace | old pp. [2] | new pp. [2]

Before (including deletions): detector

After: detector, from the You Only Look Once (YOLO) family [8], [9],

Change 88: replace | old pp. [2] | new pp. [2]

Before (including deletions): invokes

After: uses

Change 89: replace | old pp. [2] | new pp. [2]

Before (including deletions): vision-language model (VLM)

After: VLM only

Change 90: replace | old pp. [2] | new pp. [2]

Before (including deletions): reference-designator

After: reference- designator

Change 91: insert | old pp. [] | new pp. [2]

Before (including deletions): (none)

After: 2 VOLUME 11, 2023

Change 92: delete | old pp. [2] | new pp. []

Before (including deletions): confines the VLM to symbolic labeling and

After: (none)

Change 93: replace | old pp. [2] | new pp. [3]

Before (including deletions): trusting

After: from

Change 94: replace | old pp. [2] | new pp. [3]

Before (including deletions): VLM with connec- tivity, an insight we share (Section II).

After: VLM.

Change 95: replace | old pp. [2] | new pp. [3]

Before (including deletions): [5] (Auto-SPICE)

After: [3]

Change 96: replace | old pp. [2] | new pp. [3]

Before (including deletions): produce

After: gen- erate

Change 97: replace | old pp. [2] | new pp. [3]

Before (including deletions): netlists from

After: descriptions of

Change 98: replace | old pp. [2] | new pp. [3]

Before (including deletions): Most directly com- parable to our setting, the

After: The

Change 99: replace | old pp. [2] | new pp. [3]

Before (including deletions): [6] (IEEE Access, 2026)

After: [2]

Change 100: replace | old pp. [2] | new pp. [3]

Before (including deletions): a fully

After: an

Change 101: replace | old pp. [2] | new pp. [3]

Before (including deletions): image-to- SPICE

After: image- to-SPICE

Change 102: replace | old pp. [2] | new pp. [3]

Before (including deletions): circuits, pairing

After: circuits. It pairs

Change 103: replace | old pp. [2] | new pp. [3]

Before (including deletions): module (components

After: module, in which components

Change 104: delete | old pp. [2] | new pp. []

Before (including deletions): 2 VOLUME 11, 2023

After: (none)

Change 105: replace | old pp. [3] | new pp. [3]

Before (including deletions): pixels) and, as we do,

After: pixels, and it

Change 106: delete | old pp. [3] | new pp. []

Before (including deletions): automated

After: (none)

Change 107: replace | old pp. [3] | new pp. [3]

Before (including deletions): simula- tion

After: simulation

Change 108: replace | old pp. [3] | new pp. [3]

Before (including deletions): reference.

After: ref- erence.

Change 109: replace | old pp. [3] | new pp. [3]

Before (including deletions): their

After: an

Change 110: replace | old pp. [3] | new pp. [3]

Before (including deletions): they report

After: it reports

Change 111: replace | old pp. [3] | new pp. [3]

Before (including deletions): whole-netlist

After: whole- netlist

Change 112: replace | old pp. [3] | new pp. [3]

Before (including deletions): Their

After: Its

Change 113: replace | old pp. [3] | new pp. [3]

Before (including deletions): is an instance of

After: belongs to

Change 114: replace | old pp. [3] | new pp. [3]

Before (including deletions): connected-component/contour recipe

After: connected- component family that

Change 115: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: on identical wire inputs

Change 116: replace | old pp. [3] | new pp. [3]

Before (including deletions): IV-D); because their

After: V-C); its

Change 117: delete | old pp. [3] | new pp. []

Before (including deletions): (transistor-level analog, authored in-house)

After: (none)

Change 118: delete | old pp. [3] | new pp. []

Before (including deletions): (LT- spice node-voltage pass/fail)

After: (none)

Change 119: replace | old pp. [3] | new pp. [3]

Before (including deletions): ours (discrete com- ponents on CGHD-1152, component-pair F1),

After: ours, so

Change 120: replace | old pp. [3] | new pp. [3]

Before (including deletions): comparable, but the comparison shows contour/CCL node detection remains the prevailing approach, which our endpoint-graph join substantially outper- forms on a controlled, identical-input benchmark. Instance- segmentation approaches extract circuit graphs directly from pixels:

After: comparable.

Change 121: replace | old pp. [3] | new pp. [3]

Before (including deletions): [7]

After: [5]

Change 122: replace | old pp. [3] | new pp. [3]

Before (including deletions): extraction over

After: ex- traction on

Change 123: replace | old pp. [3] | new pp. [3]

Before (including deletions): in- puts, object-detection

After: inputs, object- detection

Change 124: replace | old pp. [3] | new pp. [3]

Before (including deletions): [8]

After: [6]

Change 125: replace | old pp. [3] | new pp. [3]

Before (including deletions): recent JUHCCR- v1

After: JUHCCR-v1

Change 126: replace | old pp. [3] | new pp. [3]

Before (including deletions): [9]

After: [7]

Change 127: delete | old pp. [3] | new pp. []

Before (including deletions): object

After: (none)

Change 128: delete | old pp. [3] | new pp. []

Before (including deletions): the

After: (none)

Change 129: delete | old pp. [3] | new pp. []

Before (including deletions): family [10], [11]

After: (none)

Change 130: replace | old pp. [3] | new pp. [3]

Before (including deletions): [12], with the ultimate goal of producing simulator-ready descriptions in

After: [10], and target simulator input

Change 131: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: [11]. SPICE-level modeling also supports hardware work in neighboring do- mains, including memristor-based neuromorphic circuits that realize brain-inspired learning [12] and memristive decision- making circuits for autonomous navigation

Change 132: replace | old pp. [3] | new pp. [3]

Before (including deletions): These works establish detection

After: We focus on the joining stage

Change 133: replace | old pp. [3] | new pp. [3]

Before (including deletions): recognition as largely solved; the bottleneck shifts to reconstructing electrical connectivity, the focus of our contribution.

After: evaluate it with both annotated and detected components.

Change 134: delete | old pp. [3] | new pp. []

Before (including deletions): hand- drawing

After: (none)

Change 135: replace | old pp. [3] | new pp. [3]

Before (including deletions): [1], [2]

After: [14]

Change 136: replace | old pp. [3] | new pp. [3]

Before (including deletions): annotations

After: annotations

Change 137: replace | old pp. [3] | new pp. [3]

Before (including deletions): analog/mixed-signal

After: analog/mixed-signal

Change 138: replace | old pp. [3] | new pp. [3]

Before (including deletions): [14]

After: [15]

Change 139: replace | old pp. [3] | new pp. [3]

Before (including deletions): detection, while

After: detection, and

Change 140: replace | old pp. [3] | new pp. [3]

Before (including deletions): [15] is a benchmark for evaluating

After: [16] evaluates

Change 141: replace | old pp. [3] | new pp. [3]

Before (including deletions): LLM capabilities

After: LLMs

Change 142: delete | old pp. [3] | new pp. []

Before (including deletions): Beyond circuits,

After: (none)

Change 143: replace | old pp. [3] | new pp. [3]

Before (including deletions): [16]

After: [17]

Change 144: replace | old pp. [3] | new pp. [3]

Before (including deletions): an end-to-end

After: a

Change 145: replace | old pp. [3] | new pp. [3]

Before (including deletions): diagrams,

After: diagrams,

Change 146: replace | old pp. [3] | new pp. [3]

Before (including deletions): an explicit

After: a

Change 147: replace | old pp. [3] | new pp. [3]

Before (including deletions): We use these benchmarks both as evidence of where current systems succeed (component recognition) and where they fail (connectivity), and contribute a synthetic robustness protocol that isolates the join step under controlled detector noise.

After: To our knowledge, no public benchmark reports net-level connectivity on hand- drawn circuit images; Peker et al. [2] report whole-netlist success on an in-house hand-drawn set.

Change 148: replace | old pp. [3] | new pp. [3]

Before (including deletions): [17],

After: [18]

Change 149: replace | old pp. [3] | new pp. [3]

Before (including deletions): understand- ing in particular.

After: understanding.

Change 150: replace | old pp. [3] | new pp. [3]

Before (including deletions): [18]

After: [19]

Change 151: replace | old pp. [3] | new pp. [3]

Before (including deletions): using

After: with

Change 152: delete | old pp. [3] | new pp. []

Before (including deletions): (VQA)

After: (none)

Change 153: replace | old pp. [3] | new pp. [3]

Before (including deletions): connection- matrix

After: connection-matrix

Change 154: delete | old pp. [3] | new pp. []

Before (including deletions): However, multiple independent bench- marks show that general VLMs handle component recogni- tion far better than connectivity.

After: (none)

Change 155: replace | old pp. [3] | new pp. [3]

Before (including deletions): con- nection

After: connection

Change 156: replace | old pp. [3] | new pp. [3]

Before (including deletions): [16], leading

After: [17],

Change 157: delete | old pp. [3] | new pp. []

Before (including deletions): very

After: (none)

Change 158: delete | old pp. [3] | new pp. []

Before (including deletions): at

After: (none)

Change 159: delete | old pp. [3] | new pp. []

Before (including deletions): at

After: (none)

Change 160: delete | old pp. [3] | new pp. []

Before (including deletions): and

After: (none)

Change 161: delete | old pp. [3] | new pp. []

Before (including deletions): at

After: (none)

Change 162: replace | old pp. [3] | new pp. [3]

Before (including deletions): [15]

After: [16]

Change 163: delete | old pp. [3] | new pp. []

Before (including deletions): MLLMs on

After: (none)

Change 164: replace | old pp. [3] | new pp. [3]

Before (including deletions): This corroborates the design choice, shared

After: These results vary

Change 165: replace | old pp. [3] | new pp. [3]

Before (including deletions): SINA [4], to keep wire-to-component

After: task and input, which motivates measuring VLM

Change 166: replace | old pp. [3] | new pp. [3]

Before (including deletions): out of the VLM and instead solve it with an explicit geometric model. We make this separation concrete and quantify it with

After: directly on

Change 167: replace | old pp. [3] | new pp. [3]

Before (including deletions): own Claude-VLM connectivity experiment.

After: benchmark (Section V-I).

Change 168: replace | old pp. [3] | new pp. [3]

Before (including deletions): morphological operations,

After: morphological operations

Change 169: replace | old pp. [3] | new pp. [3]

Before (including deletions): [19], typically preceded by

After: [20], usually after

Change 170: replace | old pp. [3] | new pp. [3]

Before (including deletions): threshold- ing [20]

After: thresholding [21]

Change 171: replace | old pp. [3] | new pp. [3]

Before (including deletions): [21].

After: [22].

Change 172: replace | old pp. [3] | new pp. [3]

Before (including deletions): dom- inant

After: legacy

Change 173: replace | old pp. [3] | new pp. [3]

Before (including deletions): existing pipelines

After: our pipeline

Change 174: replace | old pp. [3] | new pp. [3]

Before (including deletions): grabs all

After: takes every

Change 175: replace | old pp. [3] | new pp. [3]

Before (including deletions): pins

After: pin

Change 176: replace | old pp. [3] | new pp. [3]

Before (including deletions): (typically 30px), connected via transitive union- find [22]. This over-merges when multiple

After: (30 px), and connections are merged transitively with union-find [23]. When several

Change 177: replace | old pp. [3] | new pp. [3]

Before (including deletions): shorting

After: this rule assigns both pins of a

Change 178: replace | old pp. [3] | new pp. [3]

Before (including deletions): parts (R, C, L, D) onto

After: part to

Change 179: delete | old pp. [3] | new pp. []

Before (including deletions): SINA [4] likewise derives nets from CCL proximity, which is similarly vulnerable to over-merge under detector noise.

After: (none)

Change 180: delete | old pp. [3] | new pp. []

Before (including deletions): instead

After: (none)

Change 181: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: instead

Change 182: replace | old pp. [3] | new pp. [3]

Before (including deletions): incorporates

After: uses

Change 183: replace | old pp. [3] | new pp. [3]

Before (including deletions): geometry

After: geometry,

Change 184: replace | old pp. [3] | new pp. [3]

Before (including deletions): directional preference, and

After: we

Change 185: replace | old pp. [3] | new pp. [3]

Before (including deletions): a degree- budget completion:

After: degree-budget completion,

Change 186: replace | old pp. [3] | new pp. [3]

Before (including deletions): formulation [23], [24]

After: formula- tion [24], [25]

Change 187: replace | old pp. [3] | new pp. [3]

Before (including deletions): per-pin degree and thereby limits over-merge when endpoints are dropped or displaced. This differs from graph

After: the number of recovery connections per pin. Graph

Change 188: replace | old pp. [3] | new pp. [3]

Before (including deletions): engineering [25], which

After: engineer- ing [26]

Change 189: replace | old pp. [3] | new pp. [3]

Before (including deletions): gate-level

After: gate- level

Change 190: replace | old pp. [3] | new pp. [3]

Before (including deletions): a purely geometric model whose

After: geometric, and its

Change 191: replace | old pp. [3] | new pp. [3]

Before (including deletions): relationships

After: relations

Change 192: delete | old pp. [3] | new pp. []

Before (including deletions): wire

After: (none)

Change 193: delete | old pp. [3] | new pp. []

Before (including deletions): com- ponent

After: (none)

Change 194: delete | old pp. [3] | new pp. []

Before (including deletions): A complementary recent direction replaces rule- based connectivity with a learned model:

After: (none)

Change 195: replace | old pp. [3] | new pp. [3]

Before (including deletions): [26] pre- dicts

After: [27] predicts

Change 196: replace | old pp. [3] | new pp. [3]

Before (including deletions): schematics. Such

After: schematics; such learned

Change 197: replace | old pp. [3] | new pp. [3]

Before (including deletions): are powerful but require task-specific connec- tivity training data and, trained on printed schematics, do not transfer to hand-drawn images without retraining; our geometric model needs no

After: need

Change 198: replace | old pp. [3] | new pp. [3]

Before (including deletions): and is the focus here.

After: data, whereas our join needs none.

Change 199: replace | old pp. [3] | new pp. [3]

Before (including deletions): CircuitNet [27] likewise rely on

After: Cir- cuitNet [28] combine

Change 200: replace | old pp. [3] | new pp. [3]

Before (including deletions): plus

After: with

Change 201: delete | old pp. [3] | new pp. []

Before (including deletions): six-stage pipeline. The full pipeline proceeds through

After: (none)

Change 202: replace | old pp. [3] | new pp. [3]

Before (including deletions): detection via oriented bounding box

After: detec- tion with an oriented-bounding-box

Change 203: replace | old pp. [3] | new pp. [3]

Before (including deletions): occlusion (filling detected regions

After: occlusion, which fills each component region, enlarged by 15% of its box size (at least 5 px),

Change 204: replace | old pp. [3] | new pp. [3]

Before (including deletions): local

After: the

Change 205: replace | old pp. [3] | new pp. [3]

Before (including deletions): color);

After: color of its surroundings;

Change 206: replace | old pp. [3] | new pp. [3]

Before (including deletions): adaptive binarization;

After: binarization ($k = 0.285$, window 67 px) inside the union of component boxes plus 10 px padding;

Change 207: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: connected-component labeling after a 3×3 elliptical closing, discarding components under 28 px; (5) wire-segment and endpoint estimation by PCA of each

Change 208: replace | old pp. [3] | new pp. [3]

Before (including deletions): component labeling

After: component, followed by overlap deduplication (12■, 8 px) and an anchor filter

Change 209: replace | old pp. [3] | new pp. [3]

Before (including deletions): morphological post-processing; (5) wire end-point extraction via principal component analysis (PCA) on

After: a 16 px

Change 210: replace | old pp. [3] | new pp. [3]

Before (including deletions): regions;

After: distance;

Change 211: delete | old pp. [3] | new pp. []

Before (including deletions): wire

After: (none)

Change 212: replace | old pp. [3] | new pp. [3]

Before (including deletions): via

After: with

Change 213: replace | old pp. [3] | new pp. [3]

Before (including deletions): endpoint-graph model with degree-budget

After: endpoint graph and degree- budget

Change 214: replace | old pp. [3] | new pp. [3]

Before (including deletions): is a netlist mapping

After: maps each

Change 215: replace | old pp. [3] | new pp. [3]

Before (including deletions): pins

After: pin

Change 216: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: an

Change 217: replace | old pp. [3] | new pp. [3]

Before (including deletions): nodes, suitable for SPICE netlist generation.

After: node.

Change 218: delete | old pp. [3] | new pp. []

Before (including deletions): representative

After: (none)

Change 219: delete | old pp. [3, 4] | new pp. []

Before (including deletions): COMPONENT DETECTION Stage 1 is a YOLO26m oriented bounding box (OBB) detector fine-tuned on CGHD-1152 [1], [2]. Hand-drawn parts sit at arbitrary angles, and an axis-aligned box around a diagonal resistor also encloses the wire segments entering it, which the VOLUME 11, 2023 3 occlusion stage would then erase along with the component. The detector therefore predicts orientation. We reduce CGHD's 61 component classes to 16. Eight targets absorb variants the netlist does not distinguish. A Zener, light-emitting, or thyrector diode is a diode for the purpose of topology extraction (Table 1). Seven classes pass through unchanged: terminal, crossover, switch, text, junction, gnd, and vss. A catch-all other absorbs the remaining 30 rare classes (transformer, thyristor, triac, relay, optocoupler, and similar), which the detector localizes but does not discriminate; these devices fall outside the simulatable set and are excluded from the net-level benchmark in any case. The join model consumes component geometry and pin locations, not device identity, so class granularity finer than the netlist emits gives the detector further classes to separate while no downstream stage reads the result. Training excludes drafter_0, whose drawing style diverges markedly from the remaining drafters, leaving a random 85/15 split of 2652 training and 468 validation images. We fine-tune at 1024 px for 200 epochs with AdamW (lr0 = 10⁻³, cosine decay), batch 17, seed 42, under mosaic, mixup, RandAugment, and moderate affine and HSV jitter. On the held-out validation split the released checkpoint attains mAP@0.5 = 89.0%, mAP@0.5:0.95 = 78.5%, precision = 95.8%, and recall = 88.6%. Per-class recall ranges from 100% (operational_amplifier) down to 70.7% (crossover), which trails the next lowest class (vss, 83.1%) by 12.4 points: two wires that cross without a hop marker are locally indistinguishable from an ordinary junction-free crossing. Training logs and the full 61-to-16 class map are released with the code. C.

After: (none)

Change 220: delete | old pp. [4] | new pp. []

Before (including deletions): a set of

After: (none)

Change 221: replace | old pp. [4] | new pp. [3]

Before (including deletions): where

After: with

Change 222: replace | old pp. [4] | new pp. [3]

Before (including deletions): $\cup$ P.

After: $\cup$ P and five edge types.

Change 223: replace | old pp. [4] | new pp. [3]

Before (including deletions): with weight 0. This ensures

After: so that

Change 224: replace | old pp. [4] | new pp. [3]

Before (including deletions): of a wire are always in

After: belong to

Change 225: replace | old pp. [4] | new pp. [3]

Before (including deletions): endpoint pairs (ei, ej)

After: endpoints ei, ej on different wires

Change 226: replace | old pp. [4] | new pp. [3]

Before (including deletions): edge with weight $\blacksquare_{ei} - \blacksquare_{ej}$.

After: an edge.

Change 227: replace | old pp. [4] | new pp. [3]

Before (including deletions): captures wire

After: rule connects

Change 228: replace | old pp. [4] | new pp. [3]

Before (including deletions): corners,

After: corners

Change 229: replace | old pp. [4] | new pp. [3]

Before (including deletions): For each

After: Each

Change 230: replace | old pp. [4] | new pp. [3]

Before (including deletions): e

After: connects to at most one pin. The component-first assignment described below chooses the pin. If no component is assigned, the endpoint takes the pin within $d \leq \tau_{\text{pin}}$ with the smallest score, where the score is $d[1 - \alpha \max(0, \cos \theta)]$ with $\alpha = 0.35$

Change 231: delete | old pp. [4] | new pp. []

Before (including deletions): pin p with $\|e - p\| \leq \tau_{\text{pin}}$, add edge with weight $\|e - p\| \cdot (1 - \alpha \cdot \cos \theta)$, where

After: (none)

Change 232: delete | old pp. [4] | new pp. []

Before (including deletions): is

After: (none)

Change 233: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: outward

Change 234: replace | old pp. [4] | new pp. [3]

Before (including deletions): vector, and α controls directional preference. This

After: vector (for $d \leq 10 - 6 \cdot \text{px}$ the score

Change 235: replace | old pp. [4] | new pp. [3]

Before (including deletions): the critical

After: d). If neither step finds a pin, no

Change 236: delete | old pp. [4] | new pp. []

Before (including deletions): type: it connects wires to components. Edge type 4 (endpoint–wire–body): For each endpoint e and wire segment $w_j = (c, d)$, if the point-to-segment distance $\text{dist}(e, w_j) \leq \tau_t$, add edge with weight $\text{dist}(e, w_j)$. This captures T-junctions where a wire terminates on another wire's body. Edge type 5 (pin–wire–body): For each component pin p and wire segment w_j , if the point-to-segment distance $\text{dist}(p, w_j) \leq \tau_t$ and the projection lands mid-span (not at w_j 's endpoints, which are handled by edge type 3), add edge with weight $\text{dist}(p, w_j)$. This binds components that tap onto a passing rail along the wire's body rather than at its end. It recovers connectivity that endpoint-only binding misses, without the all-to-all over-merge of radius grabbing. The tolerances τ_{join} , τ_{pin} , and τ_t are scale-relative: $\tau = k \cdot s$ where s

After: (none)

Change 237: replace | old pp. [4] | new pp. [3]

Before (including deletions): the characteristic scale (median component diagonal), so a single parameter setting works across circuits of varying size. Component-first pin assignment: For edge type 3, we use a two-step assignment. First, assign each endpoint to its nearest component by bounding-box distance (0 if inside the bbox, using a generous radius of $\max(\tau_{\text{pin}}, 0.5 \cdot \text{diagonal})$). Second, within the assigned component, select the pin by geometric position: for horizontal components ($w > h$), left pin if endpoint $x < \text{center } x$, else right pin; for vertical components ($h > w$), top pin if endpoint $y < \text{center } y$, else bottom pin. This handles the common case where endpoints land inside a component's bounding box but far from its pin. Figure 3 contrasts the legacy radius join with our endpoint-graph model. Nets are computed as connected components of G , projected onto pins. The result is a netlist mapping each pin to a net identifier. D. DEGREE-BUDGET COMPLETION The endpoint-graph join may leave some component pins “floating”: touching only their own component's net, with no wire connecting them to another component. This indicates a dropped or over-displaced wire endpoint that the detector failed to place near the correct pin. Degree-budget completion identifies these floating pins and reconnects them via constrained optimization. A pin p is treated as floating if its net contains only pins from p 's own component, excluding inert types such as gnd, vss, text, antenna, junction, and terminal. For each floating pin p , we generate candidate edges to all pins q in other components within reach $r = p \cdot s$, where p is the reach factor; we use $p = 4$ for the deployed scale-relative join, and Section IV-D shows a broad F1 plateau for $p \in [3, 5]$. Candidate edge weight is Euclidean distance $\|p - q\|$. We then solve a min-cost b-matching problem, $\min \sum_{(p,q) \in E'} w(p, q) \cdot x_{pq}$ subject to $\sum_q x_{pq} \leq 1$ for each floating pin p , using the Hungarian algorithm [23] (scipy.optimize.linear_sum_assignment). Each target pin offers up to three slots so legitimate multi-terminal nets are not starved, and a per-pin opt-out leaves a pin with no good target unconnected rather than forcing a spurious edge. Finally, we reject any matched edge (p, q) if p and q belong to the same component or if adding the edge would merge two nets that already share a component, which structurally prevents short-circuiting two-terminal parts. Completion edges are “wireless”: they represent inferred connections without corresponding wire segments, analogous to manual pin merges in schematic editors. Figure 4 illustrates the completion step. 4

After: added.

Change 238: delete | old pp. [5] | new pp. []

Before (including deletions): TABLE 1. Class Merges: The 61 CGHD Component Classes Reduced to 16. The Eight Merged Targets Are Listed With Their Source Classes in Full; a Further Seven Classes Pass Through Unchanged (terminal, crossover, switch, text, junction, gnd, vss) and the Remaining 30 Fold Into other. Merged class n CGHD source classes resistor

After: (none)

Change 239: delete | old pp. [5] | new pp. []

Before (including deletions): resistor, resistor.adjustable, resistor.photo capacitor 3 capacitor.unpolarized, capacitor.polarized, capacitor.adjustable diode 4 diode, diode.zener, diode.light_emitting, diode.thyrector transistor 3 transistor.bjt, transistor.fet, transistor.photo inductor 3 inductor, inductor.ferrite, inductor.coupled voltage_source 3 voltage.battery, voltage.dc, voltage.ac integrated_circuit 3 integrated_circuit, integrated_circuit.ne555, integrated_circuit.voltage_regulator operational_amplifier 2 operational_amplifier, operational_amplifier.schmitt_trigger Total: 24 source classes merged into 8 targets. capacitor, transistor, and voltage_source have no bare counterpart in CGHD: they exist only as subtypes, and voltage_source renames the voltage.* family.

After: (none)

Change 240: replace | old pp. [5] | new pp. [4]

Before (including deletions): approaches compared.

After: approaches.

Change 241: replace | old pp. [5] | new pp. [4]

Before (including deletions): grabs

After: takes

Change 242: replace | old pp. [5] | new pp. [4]

Before (including deletions): causing over-merging.

After: which over-merges.

Change 243: replace | old pp. [5] | new pp. [4]

Before (including deletions): Our endpoint-graph

After: Endpoint-graph

Change 244: replace | old pp. [5] | new pp. [4]

Before (including deletions): (wire body, endpoint-pin)

After: and component-first pin assignment; directional scoring applies only to the fallback pin search. Edge type 4 (endpoint-wire-body): For an endpoint e and a different segment w_j , add an edge if the point-to-segment distance is at most τ_t and the normalized projection parameter lies in $[0.15, 0.85]$. This rule captures T-junctions away from segment ends. Edge type 5 (pin-wire-body): For a pin p and a segment w_j , add an edge if the point-to-segment distance is at most τ_t and the projection parameter lies in $[0.05, 0.95]$. This rule captures pins tapped onto a rail. The tolerances are scale-relative

Change 245: replace | old pp. [5] | new pp. [4]

Before (including deletions): directional preference resolve connectivity without over-merging.

After: fixed pixel clamps: $\tau_{\text{pin}} = \min(60, \max(24, 0.62s))$, $\tau_{\text{join}} = \min(28, \max(11, 0.30s))$ and $\tau_t = \min(20, \max(8, 0.20s))$, in pixels. Here s is the median component-box diagonal of the image (the upper middle value for an even count), with a wire-length scale as fallback when component geometry is unavailable. The multipliers and clamps are set by hand. Because s is one scalar per image, these three tolerances do not vary with component size inside a drawing. Component-first pin assignment: For edge type 3, each endpoint is first assigned to a component. Its distance to a component is 0 inside the component's OBB and otherwise the distance to the nearest OBB edge (axis-aligned box distance when OBB vertices are unavailable). The endpoint is assigned to the nearest component within $\max(\tau_{\text{pin}}, 0.5 \delta_c)$, where δ_c is that component's own box diagonal. Because this radius grows with each component's diagonal, endpoint-to-component binding follows the size of each component in addition to the image-level scale s . Within the assigned component, the endpoint takes the nearest pin. Figure 3 contrasts the legacy radius join with the endpoint graph. Nets are the connected components of G , accumulated with union-find and projected onto pins. The result maps each pin to a net identifier. C. DEGREE-BUDGET COMPLETION After base joining, some pins lie in a net that contains only pins of their own component. Completion treats these floating pins as candidates for recovery, excluding inert types (ground, supply markers, text, antennas, probes, junctions and terminals). Completion keeps the base nets and proposes connections from each floating pin p to pins q of other components in other nets. A candidate's distance must not exceed the reach $r =$

Change 246: delete | old pp. [5] | new pp. []

Before (including deletions): self-loop

After: (none)

Change 247: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: no merge between nets that already share

Change 248: replace | old pp. [5] | new pp. [4]

Before (including deletions): component's own two terminals are never matched together

After: component

Change 249: replace | old pp. [5] | new pp. [4]

Before (including deletions): After

After: after

Change 250: replace | old pp. [5] | new pp. [4]

Before (including deletions): "floating" (touching)

After: floating (connected)

Change 251: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: to

Change 252: replace | old pp. [5] | new pp. [4]

Before (including deletions): Completion

After: completion

Change 253: replace | old pp. [5] | new pp. [4]

Before (including deletions): via

After: by

Change 254: replace | old pp. [5] | new pp. [4]

Before (including deletions): dashed),

After: dashed); a guard rejects merges between nets that already share a component. $\rho_{\tau c}$, where $\tau c = \min(60, \max(24, 0.62s))$ p_x and $p = 4$. For each candidate we look for a wire witness: a detected segment whose endpoints lie near p and q in either orientation. The witness cost is the smallest, over segments and orientations, of the larger endpoint-to-pin distance, bounded by r . Without a witness, the candidate cost is $\|p - q\|$. The candidates form a rectangular assignment matrix

Change 255: replace | old pp. [5] | new pp. [4]

Before (including deletions): self-loop

After: one row per floating pin. Each target pin supplies up to three slot columns, limited by its candidate demand, and each row has a private dummy column of cost $1.5r$ for opting out. Unavailable assignments cost 106. The Hungarian algorithm [24] (`scipy.optimize.linear_sum_assignment`) assigns every row to one column, with at most one row per column; the target slots implement bounded-capacity matching [25]. Matches are applied in sequence. Dummy and unavailable assignments add no edge. A merge is skipped if both pins are already in the same net or if their nets already share a component; this

Change 256: replace | old pp. [5] | new pp. [4, 5]

Before (including deletions): preventing shorts

After: acts on completion merges only. Completion edges have no corresponding wire segment, much like a manual pin merge in a schematic editor. Figure 4 illustrates the step.

IV. EXPERIMENTAL SETUP

A. DATASETS AND IMAGE PROVENANCE All real images come from CGHD-1152 [14], a public corpus of hand-drawn circuit images organized by drafter. We use four evaluation sets. The wire benchmark has 134 images with human wire labels and annotated component boxes. The human-verified net benchmark has 31 images from 10 drafters. The 34 annotated images were chosen by circuit content, not appearance: ten initial images spanning a range of complexity, and the rest sampled at even intervals of electrical-component count from the circuits whose parts all belong to the scored device types. Image background was not a selection criterion, but the selected copies have mostly plain backgrounds, partly because 16 of them are stroke maps (see below). Three annotated images were excluded before scoring (one unverified, two failing physical-sanity checks). Net proposals were bootstrapped by a conservative endpoint-graph join over the annotated wires and corrected by one annotator in a purpose-built verification tool. The 31 images contain 3–14 electrical components each (median 7); 15 have 4 VOLUME 11, 2023 at most five and 12 have at least ten. The independent held-out benchmark (164 images, 24 drafters) and the synthetic suite (15 authored circuits) are described in Sections V-D and V-A. The 134 wire-benchmark and 31 net-benchmark images are 704×704 copies from a Roboflow export of CGHD. The export resized each original without preserving its aspect ratio. Of the 31 net-benchmark images, 14 are also flipped or rotated relative to the CGHD file, and 16 (45 of the 134 wire-benchmark images) are CGHD's binary stroke-segmentation renderings instead of photographs. Wire and net labels were drawn on these copies. The held-out benchmark uses the original CGHD photographs.

B. CONNECTIVITY METRIC For each image we form the set of unordered pairs of electrical components that share at least one net, and compare predicted with reference pairs. The electrical subset contains resistors, capacitors, inductors, diodes, transistors, sources and integrated circuits (ICs); junctions, terminals, ground symbols and crossovers affect the score only through the connections they mediate. The metric ignores which terminal of a component carries a connection. Micro-F1 pools true positives (TP), false positives (FP) and false negatives (FN) over images, so a dense board contributes in proportion to its connections; it is our primary metric. Macro-F1, the mean of per-image F1, is reported alongside.

C. STATISTICAL PROTOCOL All comparisons are paired over images. We report 95% percentile confidence intervals (CIs) from an image-level bootstrap with $B = 10,000$ resamples, and test each difference with three procedures: the bootstrap itself, a sign-flip permutation test that swaps the two methods' outputs

Change 257: replace | old pp. [5] | new pp. [5]

Before (including deletions): two-terminal components. IV.

After: each image (exact when at most 22 images differ, otherwise 100,000 Monte Carlo draws), and a Wilcoxon signed-rank test on per-image F1. The first two test the micro-F1 difference; the Wilcoxon test compares images. Holm's correction is applied within each procedure across the comparisons of a table. For the human-verified benchmark we report all three Holm-adjusted p values (Table 4); for the held-out benchmark we report the larger of the Holm-adjusted bootstrap and permutation p values (Table 5), and the Wilcoxon test is reported for the human-verified benchmark only. We also count per-image wins, ties and losses (W/T/L). Unless stated otherwise (Section V-G), every join receives annotated component boxes and the pipeline's own detected wires. V.

Change 258: delete | old pp. [5] | new pp. []

Before (including deletions): FRAMEWORK

After: (none)

Change 259: replace | old pp. [5] | new pp. [5]

Before (including deletions): author

After: authored

Change 260: delete | old pp. [5] | new pp. []

Before (including deletions): synthetic

After: (none)

Change 261: replace | old pp. [5] | new pp. [5]

Before (including deletions): spanning

After: with

Change 262: replace | old pp. [5] | new pp. [5]

Before (including deletions): loops,

After: loops

Change 263: replace | old pp. [5] | new pp. [5]

Before (including deletions): 6- component ring (cyclic and bridge topologies being the main stress cases for join robustness).

After: 6-component ring.

Change 264: replace | old pp. [5] | new pp. [5]

Before (including deletions): ground-truth netlist

After: netlist, component values

Change 265: delete | old pp. [5] | new pp. []

Before (including deletions): SPICE deck with

After: (none)

Change 266: delete | old pp. [5] | new pp. []

Before (including deletions): categories of

After: (none)

Change 267: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: types

Change 268: replace | old pp. [5] | new pp. [5]

Before (including deletions): (L0) baseline. These are

After: level L0:

Change 269: replace | old pp. [5] | new pp. [5]

Before (including deletions): with standard deviation σ (from

After: (σ from

Change 270: replace | old pp. [5] | new pp. [5]

Before (including deletions): L4),

After: L4);

Change 271: replace | old pp. [5] | new pp. [5]

Before (including deletions): corruption in which each wire is shortened inward by a fixed length to model endpoints that stop short of the pin (from 4 px at L1 to 22 px at L4), full wire drops with probability p_{drop} (up to 0.20 at L4), wrong-pin snaps with probability p_{wp} (from 0.02 at L2 to 0.10 at L4), and anchor displacement, where with probability p_{disp} one endpoint is moved by up to a_{max} pixels (from 30 px at L1 to 80 px at L4). We report

After: wires, TABLE 1. Synthetic

Change 272: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: leaderboard: mean component-pair

Change 273: replace | old pp. [5, 6] | new pp. [5]

Before (including deletions): as the component-connectivity F1 score, measuring which pairs of components share a net in the predicted versus ground-truth netlist. We report simulation accuracy as the percentage of random seeds for which every source current in the predicted SPICE netlist matches the clean oracle within 5%. B. WIRE DETECTION BENCHMARK We evaluate on 134 images from CGHD-1152 with ground-truth wire and component labels. Figure 5 and Table 2 summarize the top configurations across 36 tested variants. TABLE 2. Wire Detection Performance (134 images) Configuration F1 Prec Rec Sauvola + anchor 16 px (ours) 0.976 0.973 0.978 Sauvola + anchor 12 px 0.973 0.974 0.972 Sauvola ($w=61$) + anchor 12 px 0.959 0.944 0.974 Sauvola ($k=0.2875$) + anchor 14 px 0.950 0.921 0.980 Otsu threshold + CCL 0.789 0.796 0.783 VOLUME 11, 2023 5 0.6 0.7 0.8 0.9 1.0 Wire detection F1 (134 images) Triangle Otsu adaptive Gaussian Sauvola (alt.) Sauvola ($k=0.2875$) Sauvola ($w=61$) Sauvola, anchor 12px Sauvola, anchor 16px (ours) 0.758 0.789 0.845 0.949 0.950 0.959 0.973 0.976 FIGURE 5. Wire detection F1 across thresholding methods and pipeline variants (134 images). Our configuration (Sauvola binarization with a 16 px anchor filter) achieves $F1 = 0.976$. Otsu and triangle thresholding degrade significantly. The pipeline achieves median $F1 = 1.000$ across the benchmark set, with 87% of images scoring $F1 \geq 0.90$. Sauvola adaptive binarization dominates all alternatives: Otsu yields $F1 = 0.789$, adaptive Gaussian $F1 = 0.845$. Component occlusion matters: without it, component internals (text, symbols, internal wiring) generate hundreds of false wire detections. C. JOIN STRATEGY COMPARISON Table 3 and Figure 6 compare joining strategies on the synthetic evaluation framework (15

After: over 15

Change 274: replace | old pp. [6] | new pp. [5]

Before (including deletions): seeds, mean F1 over

After: seeds at each

Change 275: replace | old pp. [6] | new pp. [5]

Before (including deletions): levels L1–L4). TABLE 3. Join Strategy Leaderboard (Synthetic Eval)

After: level.

Change 276: replace | old pp. [6] | new pp. [5]

Before (including deletions): The completion-based strategies are

After: TABLE 2. Scale-relative completion join on eight of

Change 277: replace | old pp. [6] | new pp. [5]

Before (including deletions): most robust, main- taining

After: 15 synthetic circuits at L4 (16 seeds). Circuit Comps

Change 278: replace | old pp. [6] | new pp. [5]

Before (including deletions): ≥ 0.94

After: Prec Sim. acc. Parallel R–R 3 1.00 1.00 25% R–R divider 3 0.99 1.00 69% RL series 3 0.99 1.00 81% Diode–R loop 3 0.99 1.00 69% Dual R loops 4 0.93 0.90 12% 4-component loop 4 0.94 0.91 56% Wheatstone bridge 6 0.82 0.85 6% 6-component ring 6 0.95 0.92 50% shortened inward by 4 px

Change 279: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: L1 to 22 px at L4; wire drops with probability up to 0.20 at L4; wrong-pin snaps with probability from 0.02 at L2 to 0.10 at L4; and anchor displacement, which moves one endpoint by up to 30 px at L1 and 80 px at L4. Besides component-pair F1, we report simulation accuracy:

Change 280: replace | old pp. [6] | new pp. [5]

Before (including deletions): highest error severity (L4), compared to

After: percentage of seeds for which every source current of the predicted netlist, simulated with the authored values, matches the clean reference within 5%. Table 1 and Figure 5 compare the join strategies. At L4, the scale-relative graph with completion scores

Change 281: replace | old pp. [6] | new pp. [5]

Before (including deletions): = 0.36 for

After: 0.95,

Change 282: replace | old pp. [6] | new pp. [5]

Before (including deletions): legacy

After: rescue graph with completion 0.94, the rescue graph alone 0.90, the scale-relative graph alone 0.85 and

Change 283: replace | old pp. [6] | new pp. [5]

Before (including deletions): baseline, a 2.6× improvement.

After: 0.36.

Change 284: replace | old pp. [6] | new pp. [5]

Before (including deletions): radius join's transitive grabbing causes catastrophic over-merging under noise: at L4, it shorts multiple components onto shared nets. We emphasize, however, that the endpoint-graph base joins are themselves far more robust than the radius join, reaching F1 = 0.90 at L4 (Table 3). The degree-budget completion adds a further +0.05 over the strongest of these by recovering dropped connections the base

After: rescue

Change 285: replace | old pp. [6] | new pp. [5]

Before (including deletions): join leaves floating, while its per-pin edge budget keeps precision high. Pairing that completion with the high- precision scale-relative base (the scale-relative completion join used throughout) gives the best synthetic robustness (0.95 at L4) and, more importantly, the best real-image connectivity (Section IV-D). The large gap is against the legacy radius join; against the strongest prior graph join the improvement is smaller but consistent across every severity level. For read- ability we omit two intermediate variants whose synthetic scores fall within rounding of rows shown (nearest-pin without anchoring, and

After: differs from

Change 286: replace | old pp. [6] | new pp. [5]

Before (including deletions): base with end-extension but no dead-end rescue). D. REAL-IMAGE NET-LEVEL EVALUATION The synthetic results above measure robustness under a para- metric error model. To validate on real detector output, we constructed, to our knowledge, the first human-verified net- level ground truth for hand-drawn circuits: 31 images from CGHD-1152 in which every electrical net was traced and confirmed

After: graph

Change 287: delete | old pp. [6] | new pp. []

Before (including deletions): human annotator through a purpose-built verification tool. Because one of our comparison points is a vision-language model (Section IV-F), the human, not any model, is the verifier of record, so the answer key is independent of every method under test. Net-level proposals were bootstrapped by a conservative endpoint-graph join over the dataset's perfect wire labels and then corrected by hand; during verification the human consistently caught cases where the bootstrap join had wrongly split a single rail, and we additionally re-validate all conclusions on the independently authored synthetic ground truth to rule out residual bootstrap bias. Connectivity is scored as component-pair F1 restricted to SPICE-active components, identical to the synthetic metric. Table 4 reports every registered join strategy on the verified set, run on the pipeline's own detected wires. We report micro-averaged (pair-pooled) component-pair F1 as the primary metric, so that a dense 14-component board contributes in proportion to its connections rather than counting the same as a 3-component one; the macro (mean per-image) F1 is listed alongside for reference. The best configuration, the scale-relative completion join (degree-budget completion applied to a high-precision scale-relative endpoint-graph base), achieves micro-F1 = 0.890 (P = 0.92, R = 0.86; 95% bootstrap CI [0.855, 0.924]), improving on the prior default that applies the same completion to the rescue base (0.829) and the radius union-find join (0.667) (Figure 7). The gain over that rescue-based completion comes from the cleaner base: the rescue base's

After: (none)

Change 288: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: a

Change 289: replace | old pp. [6] | new pp. [5]

Before (including deletions): over-reach

After: step that gives dangling wire ends a longer directional reach. Table 2 breaks down our join by circuit at L4. Precision is at least 0.85

Change 290: replace | old pp. [6] | new pp. [5]

Before (including deletions): noisy detected wires

After: every circuit. The Wheatstone bridge is the hardest case, with F1 0.82

Change 291: replace | old pp. [6] | new pp. [5]

Before (including deletions): cost precision, whereas

After: simulation accuracy 6%;

Change 292: replace | old pp. [6] | new pp. [5]

Before (including deletions): scale- relative base

After: parallel R-R circuit

Change 293: delete | old pp. [6] | new pp. []

Before (including deletions): precision high while the bounded completion still recovers recall. A reach sweep shows a broad

After: (none)

Change 294: replace | old pp. [6] | new pp. [5]

Before (including deletions): plateau across reach 3–5×s,

After: 1.00 at 25% simulation accuracy,

Change 295: replace | old pp. [6] | new pp. [5]

Before (including deletions): setting is robust rather than tuned to

After: two measures capture different properties. B. WIRE DETECTION Table 3 and Figure 6 summarize the best of 36 extractor variants on the 134-image wire benchmark. Sauvola binarization with

Change 296: replace | old pp. [6] | new pp. [5]

Before (including deletions): spike. Confidence intervals are wide

After: 16 px anchor filter reaches F1 0.976 (median per-image F1 1.000; 87% of images

Change 297: replace | old pp. [6] | new pp. [5]

Before (including deletions): N = 31

After: or above 0.90). Otsu thresholding reaches 0.789

Change 298: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: adaptive Gaussian thresholding 0.845. This best variant deduplicates overlapping segments at 10■/18 px;

Change 299: replace | old pp. [6] | new pp. [5]

Before (including deletions): per-strategy intervals overlap, but

After: deployed setting, which uses 12■/8 px with

Change 300: replace | old pp. [6] | new pp. [5]

Before (including deletions): ranking is consistent

After: same anchor filter, scores 0.973

Change 301: replace | old pp. [6] | new pp. [5]

Before (including deletions): corroborated

After: the extractor used in every other experiment. The score does not depend on image modality: F1 is 0.967

Change 302: replace | old pp. [6] | new pp. [5]

Before (including deletions): independently authored synthetic suite (Table 3). This real-image ranking resolves an open question from prior work

After: 89 photographs and 0.986

Change 303: delete | old pp. [6] | new pp. []

Before (including deletions): this pipeline: against bootstrapped pilot labels

After: (none)

Change 304: replace | old pp. [6] | new pp. [5]

Before (including deletions): degree-budget completion had not consistently beaten its base join; against human-verified ground truth it clearly does (micro-F1 0.829 vs 0.787 for the rescue base alone), and the scale-relative variant is better still. The same ordering holds 6

After: 45 stroke maps.

Change 305: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: 5

Change 306: replace | old pp. [7] | new pp. [6]

Before (including deletions): 6.

After: 5.

Change 307: replace | old pp. [7] | new pp. [6]

Before (including deletions): strategy robustness

After: strategies

Change 308: delete | old pp. [7] | new pp. []

Before (including deletions): bars give

After: (none)

Change 309: replace | old pp. [7] | new pp. [6]

Before (including deletions): connectivity

After: component-pair

Change 310: delete | old pp. [7] | new pp. []

Before (including deletions): The completion-based strategies (left) stay near 1.0, while the radius union-find and nearest-pin baselines (right) collapse at L3–L4.

After: (none)

Change 311: replace | old pp. [7] | new pp. [6]

Before (including deletions): connectivity

After: component-pair

Change 312: replace | old pp. [7] | new pp. [6]

Before (including deletions): Errors cause fragmentation rather than shorts (precision stays high; Table 5); the balanced Wheatstone bridge

After: TABLE 3. Wire detection on 134 images with annotated component boxes. The deployed row uses 12■/8 px overlap deduplication and

Change 313: replace | old pp. [7] | new pp. [6]

Before (including deletions): hardest case. TABLE 4. Join Strategy and Baseline Comparison on Human-Verified Net-Level Ground Truth (31 real images, detected wires). Primary metric is micro-averaged (pair-pooled) component-pair F1; macro (mean per-image) F1 is shown for reference. Method

After: extractor used in all other experiments; the other Sauvola rows use 10■/18 px. Configuration

Change 314: replace | old pp. [7] | new pp. [6]

Before (including deletions): F1macro Deterministic joins (ours)

After: Sauvola + anchor 16 px, 12■/8 px dedup (deployed) 0.973 0.967 0.978 Sauvola + anchor 16 px (best variant) 0.976 0.973 0.978 Sauvola + anchor 12 px 0.973 0.974 0.972 Sauvola (w=61) + anchor 12 px 0.959 0.944 0.974 Sauvola (k=0.2875) + anchor 14 px 0.950 0.921 0.980 Otsu threshold + CCL 0.789 0.796 0.783 0.6 0.7 0.8 0.9 1.0 Wire detection F1 (134 images) Triangle (skeleton) Otsu adaptive Gaussian (skeleton) Sauvola (alt.) Sauvola (k=0.2875) Sauvola (w=61) Sauvola, anchor 12px Sauvola, anchor 16px (ours) 0.758 0.789 0.845 0.949 0.950 0.959 0.973 0.976 FIGURE 6. Wire detection F1 across thresholding methods

Change 315: replace | old pp. [7] | new pp. [6]

Before (including deletions): ablations) Scale-rel. graph + completion (ours) 0.890 0.919 0.864 0.901 Rescue graph + completion (prior) 0.829 0.789 0.874 0.853 Scale-rel. graph (base) 0.816 0.929 0.727 0.838 Rescue graph (base) 0.787 0.811 0.764 0.813 Radius union-find (legacy) 0.667 0.698 0.638 0.674 Classical deterministic baselines (identical detected wires) Hough + proximity (best) 0.805 0.750 0.868 0.847 Connected-components (best) 0.624 0.965 0.461 0.611 Reference (non-deterministic / oracle wires) VLM (Claude Opus 4.8) 0.923 0.970 0.880 0.949 Scale-rel. + completion, perfect wires 0.890

0.913 0.868 0.916

After: pipeline variants (134 images, annotated component boxes). The adaptive Gaussian and Triangle bars use skeleton extraction (F1 = 0.845 and 0.758). C. HUMAN-VERIFIED BENCHMARK Table 4 reports every join configuration and baseline

Change 316: replace | old pp. [7] | new pp. [6]

Before (including deletions): authored synthetic suite (Table 3), where

After: 31-image human-verified benchmark, together with

Change 317: replace | old pp. [7] | new pp. [6]

Before (including deletions): scale- relative

After: paired tests, and Fig. 7 plots the micro-F1 values with their CIs. Our join, degree-budget

Change 318: replace | old pp. [7] | new pp. [6]

Before (including deletions): join also ranks first, so

After: on

Change 319: replace | old pp. [7] | new pp. [6]

Before (including deletions): result is not an artifact of how the net-level labels were bootstrapped. The dominant connectivity recipe in prior image-to-netlist systems erases component boxes and reads nets off the connected components, or contours, of the residual wire pixels.

After: scale-relative base, reaches micro-F1 0.890 (P 0.919, R 0.864; macro- F1 0.901; TP/FP/FN 418/37/66; 95% CI [0.855, 0.924]).

Change 320: replace | old pp. [7] | new pp. [6]

Before (including deletions): is used by SINA [4], the concurrent contour-based node detector of Peker et al. [6], AMSnet [14], and the Bayer instance-segmentation pipeline [7], and was designed for clean printed schematics. On hand-drawn photographs it has no precise operating point: pencil-stroke fragmentation splits each net into many components (F1 < 0.25 at sensible morphological gap-bridging), and only pathological closing that effectively merges by proximity lifts recall, at a steep precision cost. For a controlled comparison that isolates the join algorithm from binarization quality, we run connected-component labeling

After: beats every deterministic comparator

Change 321: replace | old pp. [7] | new pp. [6]

Before (including deletions): our own detected wire segments, identified

After: every test after

Change 322: replace | old pp. [7] | new pp. [6]

Before (including deletions): connectivity

After: component-pair

Change 323: delete | old pp. [7] | new pp. []

Before (including deletions): net-level

After: (none)

Change 324: replace | old pp. [7] | new pp. [6]

Before (including deletions): (component-pair metric). Our scale-relative completion join leads all deterministic strategies

After: with annotated component boxes

Change 325: delete | old pp. [7] | new pp. []

Before (including deletions): the classical Hough and connected-component baselines (run on identical

After: (none)

Change 326: replace | old pp. [7] | new pp. [6]

Before (including deletions): wires). The dashed line marks a strong VLM (Claude Opus 4.8) evaluated on the same 31 images, from which our join is statistically indistinguishable (paired micro-F1 difference +0.033, 95% bootstrap CI [-0.009, +0.078]) while being deterministic, structurally valid by construction, and orders of magnitude cheaper.

After: wires.

Change 327: replace | old pp. [7] | new pp. [6]

Before (including deletions): Cls. tical input

After: Cls; the dashed line marks the VLM reference (Claude Opus 4.8). Holm correction. The margin over the prior default, which applies the same completion

Change 328: insert | old pp. [] | new pp. [6]

Before (including deletions): (none)

After: the rescue base, is +0.061 [+0.025, +0.100] with 11 wins, 18 ties and 2 losses. Completion lifts the scale-relative base from 0.816 to 0.890, mostly through recall (0.727 to 0.864). The scale-relative base is also more precise than the rescue base (0.929 versus 0.811), and completion keeps most of that precision (0.919, against 0.789 for completion on the rescue base). Leaving out any one image moves

Change 329: replace | old pp. [7] | new pp. [6]

Before (including deletions): join,

After: micro-F1 within 0.885–0.900,

Change 330: replace | old pp. [7] | new pp. [6]

Before (including deletions): its

After: no deletion changes the sign of any comparison. A sweep of the completion reach over $p \in [3, 5]$ gives macro-F1 0.895–0.903. Several image-to-netlist systems erase component regions and read nets from the remaining wire pixels with connected components or contours [1], [2]. On our detected wires, the

Change 331: insert | old pp. [] | new pp. [6]

Before (including deletions): (none)

After: connected-component

Change 332: delete | old pp. [7] | new pp. []

Before (including deletions): only

After: (none)

Change 333: delete | old pp. [7] | new pp. []

Before (including deletions): =

After: (none)

Change 334: insert | old pp. [] | new pp. [6]

Before (including deletions): (none)

After: at precision 0.965: it rarely merges wrongly, but every stroke gap splits a net

Change 335: replace | old pp. [7] | new pp. [6, 7]

Before (including deletions): collapses to 0.46:

After: 0.461). Hough line fitting [20] 6 VOLUME 11, 2023 TABLE 4. Component-pair connectivity on

Change 336: replace | old pp. [7] | new pp. [7]

Before (including deletions): blobs short or drop most nets), versus 0.890

After: 31-image human-verified benchmark (annotated component boxes; detected wires unless noted). F1, P and R are micro-averaged. ■ is the micro-F1 of ours minus the comparator with 95% paired bootstrap CI; Holm-adjusted p values are given

Change 337: replace | old pp. [7] | new pp. [7]

Before (including deletions): endpoint-graph join (Table 4). A stronger classical alternative, Hough-transform line detection [19] followed by

After: bootstrap, the permutation test on micro-F1 and the Wilcoxon test on per-image F1; W/T/L counts per-image wins, ties and losses from our side. Method F1 P R F1macro Δ [95% CI] Holm p (boot / perm / Wilcoxon) W/T/L Endpoint-graph joins Scale-rel. graph + completion (ours) 0.890 0.919 0.864 0.901 — — — Rescue graph + completion 0.829 0.789 0.874 0.853 +0.061 [+0.025, +0.100] <0.001 / 0.001 / 0.009 11/18/2 Scale-rel. graph (base) 0.816 0.929 0.727 0.838 +0.075 [+0.038, +0.113] <0.001 / 0.002 / 0.005 14/16/1 Rescue graph (base) 0.787 0.811 0.764 0.813 +0.103 [+0.065, +0.142] <0.001 / <0.001 / 0.002 17/14/0 Baselines on the same detected wires Radius union-find (legacy) 0.667 0.698 0.638 0.674 +0.224 [+0.166, +0.281] <0.001 / <0.001 / <0.001 25/6/0 Hough +

Change 338: replace | old pp. [7] | new pp. [7]

Before (including deletions): linking, fares better, since line fitting denoises the fragmented strokes, and

After: (best) 0.805 0.750 0.868 0.847 +0.086 [+0.045, +0.127] <0.001 / 0.002 / 0.030 19/8/4 Connected components (best) 0.624 0.965 0.461 0.611 +0.267 [+0.136, +0.409] <0.001 / <0.001 / <0.001 21/8/2 References Ours with annotated wires 0.890 0.913 0.868 0.916 — — — VLM (Claude Opus 4.8) 0.923 0.970 0.880 0.949 -0.033 [-0.078, +0.008] 0.12 / 0.15 / 0.030 7/11/13 with proximity linking

Change 339: delete | old pp. [7] | new pp. []

Before (including deletions): micro-F1 =

After: (none)

Change 340: replace | old pp. [7] | new pp. [7]

Before (including deletions): best-tuned

After: best

Change 341: replace | old pp. [7] | new pp. [7]

Before (including deletions): tolerance, but still trails our join by 0.085 at markedly lower precision. Each baseline is reported at its

After: tolerance. Both baselines use the

Change 342: replace | old pp. [7] | new pp. [7]

Before (including deletions): over a tolerance sweep, so

After: of their sweep on

Change 343: replace | old pp. [7, 8] | new pp. [7]

Before (including deletions): are upper bounds on the classical recipes. Explicit typed-endpoint reasoning with bounded completion thus outperforms both blob/contour connectivity and line- VOLUME 11, 2023 7 primitive proximity tracing.

After: 31 images, which favors them.

Change 344: replace | old pp. [8] | new pp. [7]

Before (including deletions): additionally attempted

After: also tried

Change 345: replace | old pp. [8] | new pp. [7]

Before (including deletions): image-to- netlist

After: image-to-netlist

Change 346: delete | old pp. [8] | new pp. []

Before (including deletions): end-to-end

After: (none)

Change 347: replace | old pp. [8] | new pp. [7]

Before (including deletions): our hand-drawn benchmark, and none could be fairly applied.

After: the benchmark.

Change 348: replace | old pp. [8] | new pp. [7]

Before (including deletions): classical printed- schematic

After: printed-schematic

Change 349: replace | old pp. [8] | new pp. [7]

Before (including deletions): [3] installs and

After: [4]

Change 350: replace | old pp. [8] | new pp. [7]

Before (including deletions): connectivity, a domain- transfer failure.

After: con- nectivity.

Change 351: replace | old pp. [8] | new pp. [7]

Before (including deletions): [26]

After: [27]

Change 352: delete | old pp. [8] | new pp. []

Before (including deletions): which we ran end-to-end with the released model,

After: (none)

Change 353: replace | old pp. [8] | new pp. [7]

Before (including deletions): active

After: released

Change 354: replace | old pp. [8] | new pp. [7]

Before (including deletions): inductor,

After: inductor

Change 355: insert | old pp. [] | new pp. [7]

Before (including deletions): (none)

After: in

Change 356: replace | old pp. [8] | new pp. [7]

Before (including deletions): circuits contain

After: circuits,

Change 357: replace | old pp. [8] | new pp. [7]

Before (including deletions): appearance, so it cannot represent hand-drawn discrete circuits.

After: drawings.

Change 358: replace | old pp. [8] | new pp. [7]

Before (including deletions): [5]

After: [3]

Change 359: replace | old pp. [8] | new pp. [7]

Before (including deletions): equivalent in spirit to

After: which

Change 360: replace | old pp. [8] | new pp. [7]

Before (including deletions): experiment in Section IV-F, rather than a deterministic algorithm. SINA's [4] anonymized

After: reference (Section V-I) covers. The

Change 361: replace | old pp. [8] | new pp. [7]

Before (including deletions): was unextractable

After: of SINA [1] could not be extracted,

Change 362: replace | old pp. [8] | new pp. [7]

Before (including deletions): [27]

After: [28]

Change 363: delete | old pp. [8] | new pp. []

Before (including deletions): only Co-lab

After: (none)

Change 364: replace | old pp. [8] | new pp. [7]

Before (including deletions): released

After: trained

Change 365: replace | old pp. [8] | new pp. [7]

Before (including deletions): This survey is itself informative: no prior system provides a deterministic, hand-drawn-ready connectivity module, so the controlled

After: The

Change 366: replace | old pp. [8] | new pp. [7]

Before (including deletions): above,

After: therefore stand in for these approaches on identical inputs. Replacing the detected wires with the annotated wires, with boxes held fixed, changes micro-F1 from 0.8903 to 0.8898 and macro-F1 from 0.901 to 0.916. On this benchmark, the extracted wires are as good as the annotated ones for joining. The join also scores alike on

Change 367: replace | old pp. [8] | new pp. [7]

Before (including deletions): run

After: image modalities: micro-F1 is 0.883

Change 368: insert | old pp. [] | new pp. [7]

Before (including deletions): (none)

After: the 15 photographs and 0.895 on the 16 stroke maps. D. INDEPENDENT HELD-OUT BENCHMARK The 31-image benchmark has three weaknesses: it is small, its labels were bootstrapped from a graph join similar to ours, and

Change 369: replace | old pp. [8] | new pp. [7]

Before (including deletions): identical detected wires,

After: configuration was chosen on it. We therefore built a second benchmark whose labels come from the CGHD authors' own annotations and whose images were not used to develop or tune the join. The extractor configuration was selected earlier on the wire benchmark, whose resized copies include 25 of the held-out images. Excluding those 25 images (139 remain) gives micro-F1 0.709, and excluding every photograph of their drawings (123 remain) gives 0.704; in both cases our join still ranks first, with every margin still significant. Reference construction. CGHD provides binary stroke-segmentation maps and symbol instance polygons for a subset of its images. Extending the dataset's own wire-generation tooling, we subtract the dilated symbol polygons from the stroke map and take the remaining connected components as conductors. A conductor connects every polygon it touches, except text; junction polygons merge the conductors they touch. At a crossover polygon, the arms are paired straight through by their local direction ($180^\circ \pm 45^\circ$), and a crossover that cannot be paired is flagged as ambiguous. Composite outlines (transformers, optocouplers and relays that contain sub-parts) group their parts but take no contacts. Of 257 images, 179 pass all quality checks. Of the other 78, 16 contain no pair of electrical components and 62 fail at least one check: an ambiguous crossover (29 images), an isolated part (17) or an over-connected part (28), with 10 images failing more than one. Validation. On the 17 images shared with the human-verified benchmark, the reference agrees with the human nets at micro-F1 0.939 (P 0.993, R 0.891; exact on 11 of 17; 0.964 on the 15 clean images). We inspected every disagreement. Most are pairs the reference misses because of stroke gaps or ambiguous crossings, and seven follow from a convention: the human nets bridge switches. In five pairs, the reference was right and the human labels wrong. The reference mainly under-connects, so it understates the precision of every method. We fixed the quality filter and a junction-handling rule after inspecting these 17 overlaps. Protocol. The held-out set contains the 164 clean images outside the 31, with 3874 reference pairs from 24 drafters (all except drafter_0, which the detector's training also excluded). The input is the CGHD photograph with its aspect ratio preserved and its long side resized to 1024 px. Component boxes

Change 370: replace | old pp. [8] | new pp. [7]

Before (including deletions): fair points

After: CGHD symbol annotations, the same annotations the reference uses. We chose this resolution on wire-level labels

Change 371: replace | old pp. [8] | new pp. [7, 8]

Before (including deletions): comparison,

After: the 52 wire-benchmark overlaps, before computing any join score. Unlike the 31-image benchmark, which kept only circuits whose parts all belong to the scored device types, the held-out set also contains parts outside the scored types, most VOLUME 11, 2023 7 TABLE 5. Independent held-out benchmark (164 CGHD photographs, 24 drafters, 3874 reference pairs; annotated component boxes). ■ is ours minus the comparator in micro-F1, with 95% paired bootstrap CI. Every difference has Holm-adjusted $p = 0.0006$, the resolution floor of the tests. Method F1 P R Δ [95% CI] W/T/L Ours 0.711 0.766 0.664 – – Rescue + compl. 0.659 0.622 0.699 +0.053 [+0.033, +0.074] 85/44/35 Scale-rel. base 0.595 0.785 0.479 +0.117 [+0.097, +0.138] 112/46/6 Rescue base 0.592 0.631 0.558 +0.119 [+0.097, +0.142] 113/31/20 Radius union-find 0.518 0.478 0.565 +0.194 [+0.159, +0.229] 136/13/15 Hough 0.480 0.328 0.895 +0.231 [+0.168, +0.282] 90/22/52 Conn. comp. 0.594 0.798 0.473 +0.117 [+0.083, +0.156] 117/12/35 often switches and potentiometers (30 images each), speakers (21), transformers (16), thyristors (13) and logic gates (10); 94 of the 164 images contain at least one. These parts take part in the topology but are not scored. Results. Table 5 gives the results. Our join reaches micro-F1 0.711 [0.671, 0.748] (P 0.766, R 0.664; macro-F1 0.724) and beats every comparator, with the same ordering of the endpoint-graph variants as on the human-verified set. Hough linking keeps recall high (0.895) but merges many nets (precision 0.328); the best Hough configuration tuned on this set itself reaches 0.509. The result holds under two changes to the evaluation set: excluding all other photographs of the 31 benchmark drawings (155 images) gives 0.707, and scoring all 257 images including the unclean ones gives 0.626. Across the 24 drafters, our micro-F1 ranges from 0.465 to 0.917 (median 0.709, interquartile range 0.658–0.770),

Change 372: replace | old pp. [8] | new pp. [8]

Before (including deletions): exceeds both. Running

After: is

Change 373: insert | old pp. [] | new pp. [8]

Before (including deletions): (none)

After: of the evaluated methods for 15 drafters. Scoring conventions. The primary scoring follows the human-verified benchmark: it scores only the electrical device types, treats switches as components that are not scored, and lets separate ground symbols

Change 374: insert | old pp. [] | new pp. [8]

Before (including deletions): (none)

After: nodes only through drawn wires. We rescored the same predictions under three alternatives, applied symmetrically to reference and predictions. Adding switches, potentiometers, photoresistors, fuses, lamps, varistors, crystals, speakers and microphones to the scored set (577 more reference pairs in 76 images) gives 0.709, with a margin of +0.055 [+0.036, +0.077] over the rescue-graph completion. Treating every switch as closed gives 0.712 (+0.052). Merging all ground symbols into one node, and all negative-supply symbols into another, adds 778 pairs in 43 images and gives 0.679 (+0.035 [+0.012, +0.059]); the merged ground net concentrates many pairs, so one missed ground attachment costs several. With all three changes combined, our join scores 0.676 and leads every baseline, with Holm-adjusted $p \leq 0.009$. Human audit. A human annotator checked the derived nets of 22 of 40 held-out images drawn at random from the 164, in the verification tool used for the human-verified benchmark, correcting any errors. The tool lists images from the fewest to the most electrical components, so the 22 checked are mostly the smaller circuits: median 6 electrical components, against 16 for the 18 unchecked. Against the corrected nets, the reference scores micro-F1 0.998 (precision 1.000, recall 0.996; 283 pairs): it was exact on 21 images and missed one pair. Our join scores 0.803 against the corrected nets and 0.801 against the reference on these images, and it leads the rescue-graph completion by +0.054 [+0.018, +0.087], Hough linking by +0.101 and connected components by +0.196. The annotator verified proposed nets rather than labeling from scratch, so the audit measures how many corrections the reference needs; checking one image took a median of 77 s (mean 123 s) over 20 timed saves. Gap to the human-verified benchmark. The drop from 0.890 to 0.711 comes mainly from the input images. The held-out photographs belong to the CGHD subset for which the dataset provides stroke maps, and for this subset the 704×704 benchmark copies are mostly those stroke maps (16 of the 17 overlaps). Many of these photographs show grid paper or cluttered backgrounds, which the 31-image copies rarely contain; on them, wire F1 falls to about 0.80–0.82, against 0.98

Change 375: replace | old pp. [8] | new pp. [8, 9]

Before (including deletions): dataset's perfect

After: benchmark copies. On the 17 overlapping images, with our annotated boxes and the human nets, our join scores 0.896 on the benchmark copies and 0.728 on the photographs at long side 1024. Re-weighting the held-out images to the circuit-size distribution of the 31 raises our score by about 0.01. The stored held-out predictions for the same 17 photographs, which use the CGHD symbol boxes, score 0.714 against the human nets and 0.723 against the reference, so the choice of labels changes the score by about 0.01; on these 17 photographs Hough linking scores higher than our join (0.788 against the human nets, 0.738 against the reference). Input quality explains much of the rest, but not all of it: with the CGHD stroke maps as input instead of the photographs, our held-out score rises only to 0.775. The residual is consistent with some optimism in the 31-image score, on which our configuration was selected. Across both benchmarks, our join ranks first and the endpoint-graph variants keep their order; the classical baselines do not keep theirs (Hough ranks below connected components on the held-out set). E. ROBUSTNESS 1) Circuit complexity Table 6 splits both benchmarks by the number of electrical components. On the 31 images, F1 falls with complexity for every method, including the VLM: across images, our F1 correlates with component count at Spearman $\rho = -0.53$ ($p = 0.002$). Our recall drop from the ≤ 5 to the ≥ 10 stratum, -0.068 [-0.18 , $+0.05$] (permutation $p = 0.32$), is not significant. On images with at least ten components, our join beats the rescue-graph completion, the scale-relative base and Hough by +0.078 to +0.106, with every CI excluding zero, and trails the VLM by -0.026 [-0.087 , $+0.030$]. On the held-out benchmark, micro-F1 falls from 0.845 for at most five components to 0.693 for 20 or more, while Hough falls to 0.347. 2) Component-size variation within an image A single scale s per image could fail when component sizes vary within a drawing. On the 31 images, the overall dis- 8 VOLUME 11, 2023 TABLE 6. Micro-F1 by number of electrical components per image. The VLM was run on the human-verified benchmark only. Components Images Ours Rescue + compl. Hough VLM Human-verified benchmark (31 images) ≤ 5 15 0.940 0.933 0.892 0.982 6–9 4 0.846 0.817 0.843 0.902 ≥ 10 12 0.886 0.808 0.779 0.912 Held-out benchmark (164 images) ≤ 5 41 0.845 0.856 0.835 – 6–9 41 0.779 0.756 0.701 – 10–19 60 0.694 0.656 0.593 – ≥ 20 22 0.693 0.608 0.347 – perison of component sizes is unrelated to F1 ($\rho = +0.18$, $p = 0.33$). The ratio of the largest to the smallest electrical- component diagonal correlates with F1 at $\rho = -0.60$, but it also grows with component count ($\rho = 0.62$); controlling for count, the partial correlation is -0.41 ($p = 0.024$) for our join and -0.34 ($p = 0.064$) for the VLM. The size of the largest component relative to s has partial correlation near zero with F1. The three images that combine an IC with small discrete parts score 0.854 against 0.898 on the other 28. 3) Controlled rescaling We resampled all 31 images by factors from 0.35 to 3 and re-projected the annotated boxes and wires (Fig. 8). With annotated wires, our join stays within ± 0.005 of its native score from $0.5\times$ to $3\times$, with every CI including zero. It drops only at $0.35\times$ (-0.035 [-0.061 , -0.005]), where the lower pixel clamps bind for every image. With annotated wires, radius union-find with its fixed 30 px radius falls from 0.690 at native scale to 0.429 at $2\times$ and 0.153 at $3\times$. With wires re- extracted from each rescaled image, the full pipeline scores 0.580, 0.690, 0.811, 0.890, 0.877, 0.866 and 0.866 at 0.35, 0.5, 0.75, 1, 1.5, 2 and $3\times$. Upscaling costs at most 0.024 (CIs include zero). The loss under downscaling comes from

Change 376: replace | old pp. [8] | new pp. [9]

Before (including deletions): labels rather

After: extraction, whose Sauvola window, minimum component area and occlusion margins are set in pixels: at 0.35× the extractor finds 241 of 675 annotated segments. Completion recovers much of that loss; the scale-relative base alone scores 0.21 at 0.35×. To separate the contribution of the scale-relative tolerances, we ran the same algorithm with fixed pixel tolerances (τ_{pin} , τ_{join} , τ of 30, 14 and 10 px, completion reach 120 px). It stays within 0.016 of our join between 0.5× and 2×. The scale-relative version wins only where the fixed values are most mismatched: by +0.015 with annotated wires at 0.5× and by +0.033 in the full pipeline at 3×, both with CIs excluding zero. The fixed-pixel variant is ahead at 1.5× with annotated wires (−0.012 [−0.025, −0.001]) and at 0.75× in the full pipeline (−0.016 [−0.034, −0.001]). Most of the scale robustness therefore comes from component-relative endpoint assignment and completion; the scalar tolerances $\tau = k \cdot s$ contribute a smaller share. The clamps are tuned to native resolution: $k \cdot s$ falls outside a clamp for 39% of 0.0 0.2 0.4 0.6 0.8 1.0 Connectivity micro-F1 (a) Join only: annotated wires 0.35 0.5 0.75 1 1.5 2 3 Image scale factor f (log scale; native = 1) 0.0 0.2 0.4 0.6 0.8 1.0 Connectivity micro-F1 (b) Full: wires re-extracted 95% bootstrap CI Scale-rel. graph + completion Fixed-px graph + completion Scale-rel. graph (base) Radius union-find (30 px) FIGURE 8. Component-pair micro-F1 on the 31 images resampled by factor f (native $f = 1$). (a) Join only, with annotated wires rescaled with the image. (b) Full pipeline, with wires re-extracted from each rescaled image. Components are annotated boxes in both panels; shaded bands are 95% bootstrap CIs for our join. images at native scale, and removing the clamps costs 0.023 in the full pipeline at native scale. 4) Synthetic mixed-size circuits We redrew one component per synthetic circuit at 0.5, 2 or 3 times its size, keeping only geometrically clean variants, and applied the L1–L4 error model. Enlarging one component costs our join about 0.005 in mean F1, and it stays about 0.03 above the fixed-pixel variant. Shrinking one component costs 0.012. The failure case is a Wheatstone bridge with one 40×15 px arm: in all five variants, the scale-relative methods and radius union-find add one spurious pair even on clean wires, while the fixed-pixel graphs stay exact. A component much smaller

Change 377: replace | old pp. [8] | new pp. [9]

Before (including deletions): detected wires

After: the median is thus the weak point of a median-derived tolerance. F. ABLATIONS Table 7 removes individual edge rules from the base graph, with and without completion. Disabling T-junctions, rail taps or directional preference

Change 378: insert | old pp. [] | new pp. [9]

Before (including deletions): (none)

After: the aggregate base-graph counts unchanged, so these rules give no measurable gain on this benchmark. Fixed-pixel base tolerances raise base

Change 379: replace | old pp. [8] | new pp. [9]

Before (including deletions): unchanged

After: from 0.816 to 0.820. With completion, every row ties

Change 380: replace | old pp. [8] | new pp. [9, 10]

Before (including deletions): 0.890,

After: TP/FP/FN 418/37/66. The rescaling and mixed-size experiments (Section V-E) test the tolerance design under conditions that this benchmark does not cover. VOLUME 11, 2023 9 TABLE 7. Edge-rule ablation on the 31-image benchmark, for the base graph and the full pipeline with completion. Base graph Full pipeline Configuration F1 TP/FP/FN F1 TP/FP/FN All edge types (ours) 0.816 352/27/132 0.890 418/37/66 No T-junctions 0.816 352/27/132 0.890 418/37/66 No rail taps 0.816 352/27/132 0.890 418/37/66 No T-junctions, no rail taps 0.816 352/27/132 0.890 418/37/66 No directional preference 0.816 352/27/132 0.890 418/37/66 Fixed-pixel base tolerances 0.820 356/28/128 0.890 418/37/66 TABLE 8. Mechanism ablations on the 31-image benchmark. “Changed” counts images whose predicted pairs differ from the reference arm. Arm TP/FP/FN F1 F1macro Changed Occlusion on (ours) 418/37/66 0.890 0.901 – Occlusion off 103/26/381 0.336 0.294 31/31 Guard on (ours) 418/37/66 0.890 0.901 – Guard off 418/37/66 0.890 0.901 0/31 Witness required 413/37/71 0.884 0.907 6/31 Table 8 tests the remaining mechanisms. Removing only the occlusion step, with cropping and all extractor settings held fixed, exposes component pixels to the extractor, which then returns 466 wire segments instead of 666; micro-F1 falls from 0.890 to 0.336 (P 0.798, R 0.213), and predictions change on all 31 images. The shared-component guard fires on 18 of 31 images and rejects 38 candidate merges. These merges would join two pins of the same component, which the component-pair metric cannot see, so the scores are unchanged. Requiring a wire witness for every completion edge changes six images and lowers micro-F1 to 0.884

Change 381: replace | old pp. [8] | new pp. [10]

Before (including deletions): only

After: to 0.907. The rescaling experiment adds two further arms, fixed-pixel tolerances and unclamped tolerances (Section V-E), and Section V-G decomposes the loss

Change 382: replace | old pp. [8] | new pp. [10]

Before (including deletions): 0.901 to 0.916 (Table 4), so

After: detected components. G. END-TO-END WITH DETECTED COMPONENTS We replaced

Change 383: replace | old pp. [8] | new pp. [10]

Before (including deletions): wire detector accounts for essentially zero pair-weighted F1. The residual error is intrinsic join ambiguity on hand-drawn geometry, not wire detection, consistent

After: annotated boxes

Change 384: replace | old pp. [8] | new pp. [10]

Before (including deletions): 0.976 wire-detection

After: outputs of our trained component detector, a YOLO26m-OBB model with 16 merged classes that reaches a mean average precision (mAP@0.5) of 89.0% on its validation split, with its lowest class recall (70.7%) on crossovers. The detector runs on the CGHD original, corrected for its Exchangeable Image File Format (EXIF) orientation tag, at its training size (long side 1024 px), and its boxes are mapped into the benchmark frame. We use the deployment confidence threshold of 0.5 without tuning. On the electrical components, detection precision, recall and

Change 385: replace | old pp. [8] | new pp. [10]

Before (including deletions): reported above (Table 2).

After: are 0.886, 0.858 and 0.872, and the same when the class must also match. As a check of the scoring path, feeding the annotated boxes through the same mapping and matching code reproduces 418/37/66. Table 9 gives the results. End-to-end micro-F1 is 0.627 [0.519, 0.731] (macro-F1 0.656), a change of -0.263 [-0.366 , -0.170] from annotated boxes; the lower thresh- TABLE 9. End-to-end joining with detected components on the 31-image benchmark.

Change 386: replace | old pp. [8] | new pp. [10]

Before (including deletions): join degrades only gently

After: lower block replaces annotated boxes

Change 387: replace | old pp. [8] | new pp. [10]

Before (including deletions): circuit size,

After: detector output one factor at a time (confidence 0.5); ■ is the change from the previous row. Components F1 Prec Rec TP/FP/FN Annotated boxes 0.890 0.919 0.864 418/37/66 Detector, conf 0.25 0.613 0.575 0.655 317/234/167 Detector, conf 0.35 0.616 0.582 0.655 317/228/167 Detector, conf 0.50 0.627 0.603 0.653 316/208/168 Sequential step F1 Δ TP/FP/FN Drop missed components 0.759 -0.131 329/54/155 Add spurious detections 0.687 -0.072 322/131/162 Use detected geometry 0.644 -0.043 329/208/155 Use detected classes 0.627 -0.017 316/208/168 olds give 0.613 and 0.616. Replacing the annotated boxes one factor at a time attributes -0.131 to missed electrical components, -0.072 to spurious detections, -0.043 to box localization and -0.017 to classification;

Change 388: replace | old pp. [8] | new pp. [10]

Before (including deletions): per-image F1 falling from 0.94

After: a sequential decomposition, the shares depend

Change 389: replace | old pp. [8] | new pp. [10]

Before (including deletions): circuits with ≤ 6

After: the order of the steps. Missed

Change 390: insert | old pp. [] | new pp. [10]

Before (including deletions): (none)

After: cost the most because every pair involving a missed component becomes a false negative. The detector is sensitive

Change 391: replace | old pp. [8] | new pp. [10]

Before (including deletions): 0.86 above that and only weak correlation $r = -0.19$, and remains high-precision throughout, in contrast

After: orientation (its training used rotation augmentation of at most $\pm 10^\circ$). On the 14 images whose benchmark copy is re-oriented relative

Change 392: replace | old pp. [8] | new pp. [10]

Before (including deletions): VLM, whose precision collapses

After: CGHD file, end-to-end micro-F1 is 0.491, against 0.739

Change 393: replace | old pp. [8] | new pp. [10]

Before (including deletions): densest circuits (Section IV-F). E. PER-CIRCUIT ANALYSIS Table

After: other 17. H. CROSSOVERS Edge types 2, 4 and

Change 394: replace | old pp. [8] | new pp. [10]

Before (including deletions): shows

After: do not consult crossover labels, so a missed crossover could merge two independent wires. The 31 images contain 13 annotated crossovers in eight images. With annotated boxes, recall on these eight images is 0.834, against 0.884 on

Change 395: replace | old pp. [8] | new pp. [10]

Before (including deletions): scale-relative completion join's performance across eight representative circuits

After: 23 without a crossover ($p = 0.51$). In C242_D1_P1, suppressing the two endpoint–endpoint edges adjacent to crossings changes TP/FP/FN

Change 396: insert | old pp. [] | new pp. [10]

Before (including deletions): (none)

After: 27/4/0 to 19/0/8:

Change 397: replace | old pp. [8] | new pp. [10]

Before (including deletions): 15-circuit synthetic corpus at L4 (extreme error injection, 16 seeds). TABLE 5. Scale-Relative Completion Join Performance per Circuit at L4 (16 seeds) Circuit Comps F1 Prec Sim. acc. Parallel R–R 3 1.00 1.00 25% R–R divider 3 0.99 1.00 69% RL series 3 0.99 1.00 81% Diode–R loop 3 0.99 1.00 69% Dual R loops 4 0.93 0.90 12% 4-component loop 4 0.94 0.91 56% Wheatstone bridge

After: edges that create the false pairs also carry true connections. Deleting the annotated crossover boxes before joining changes predictions on

Change 398: replace | old pp. [8] | new pp. [10]

Before (including deletions): 0.82 0.85 6% 6-component ring 6 0.95 0.92 50% Precision remains high (≥ 0.85) across all circuits even at L4: errors cause fragmentation (dropped connections) rather than shorts (false merges). The Wheatstone bridge is

After: of

Change 399: replace | old pp. [8] | new pp. [10]

Before (including deletions): hardest topology ($F1 = 0.82$): its balanced arms make

After: 8 crossover images and raises micro-F1 by 0.010, so on annotated boxes

Change 400: replace | old pp. [8] | new pp. [10]

Before (including deletions): inferred node structure sensitive to a single mis-placed connection, and simulation accuracy falls to 6%. F. VLM CONNECTIVITY EXPERIMENT To quantify the separation argued in Section II, that connectivity is better solved by an explicit geometric model than left to a VLM, we ran a first-party experiment with a strong general VLM (Claude Opus 4.8). To make the comparison fair, the VLM is given exactly what

After: crossover labels do not help

Change 401: replace | old pp. [8] | new pp. [10]

Before (including deletions): receives:

After: overall. In

Change 402: replace | old pp. [8] | new pp. [10]

Before (including deletions): scanned

After: end-to-end run, the detector finds all 13 crossovers and labels each as a crossover at every tested threshold, so none of the false pairs in Table 9 comes from a missed or misclassified crossover; removing the detected crossovers lowers end-to-end micro- F1 by 0.019. These images were probably in the detector's training split, so this is not a held-out estimate of crossover recall, and thirteen instances are too few to test the 70.7% validation recall. I. VLM REFERENCE We ran Claude Opus 4.8 as a reference system. It receives the annotated boxes of the electrical components, drawn on the

Change 403: replace | old pp. [8] | new pp. [10]

Before (including deletions): with

After: and numbered without class labels. Our join receives

Change 404: delete | old pp. [8] | new pp. []

Before (including deletions): ground-truth component

After: (none)

Change 405: replace | old pp. [8] | new pp. [10, 11]

Before (including deletions): labelled

After: with their classes, plus the annotated junction, terminal, ground and crossover boxes. Neither method receives annotated wires. The VLM was run on the human- 10 VOLUME 11, 2023 TABLE 10. Micro-F1 of our join

Change 406: replace | old pp. [8] | new pp. [11]

Before (including deletions): index, and is asked to return, as JSON,

After: CGHD drafter on

Change 407: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: 31-image benchmark (drafters with at least four images). Drafter Images F1 F1macro TP/FP/FN drafter_1 7 0.933 0.907 56/6/2 drafter_2 5 0.854 0.874 85/14/15 drafter_10 5 0.987 0.995 37/1/0 drafter_12 4 0.913 0.938 21/0/4 All 31 0.890 0.901 418/37/66 verified benchmark only. The model returns JSON

Change 408: replace | old pp. [8] | new pp. [11]

Before (including deletions): we

After: each image is a separate call with no shared context. We

Change 409: replace | old pp. [8] | new pp. [11]

Before (including deletions): the resulting component-pairs

After: its output

Change 410: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: component-pair

Change 411: replace | old pp. [8] | new pp. [11]

Before (including deletions): Each

After: The VLM reaches micro-F1 0.923 (P 0.970, R 0.880; macro-F1 0.949) and is exact on 21 of 31 images, against 11 for our join. On pooled pairs, our join trails by -0.033 $[-0.078, +0.008]$, which is not significant (Holm-adjusted bootstrap $p = 0.12$, permutation $p = 0.15$; Table 4). Per image, the VLM is better: it wins 13 images, ties 11 and loses 7 (Wilcoxon $p = 0.030$). Its errors follow the same complexity trend as ours (Table 6). The two systems differ in what they require: the VLM needs a call to a proprietary API for every

Change 412: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: and returns no geometric trace of its decisions, whereas our join runs locally and deterministically without connectivity training data, and every recovered connection maps to a wire segment or a completion edge. J. PER-DRAFTER RESULTS Table 10 groups the 31-image results by CGHD drafter, using the drafter paths of the published CGHD-1152 file index, which place every benchmark filename under exactly one drafter. The four drafters with at least four images range from micro-F1 0.854 to 0.987. The held-out benchmark gives the spread over all 24 drafters (Section V-D). VI. DISCUSSION The endpoint-graph join with completion is the strongest deterministic method on both real benchmarks, and its margins are significant on both, which differ in labels and images and share part of their drafters. The held-out benchmark addresses the three main weaknesses of the human-verified set: its labels come from the dataset authors' annotations and not from a join, it covers 24 of the 25 drafters, and its images were not used to develop the join. Completion accounts for most of the gain over the base graphs, and on the high-precision scale- relative base it adds recall while keeping precision above 0.9 on the human-verified set. The experiments also locate the remaining errors. With annotated components on clean copies, extracted wires are as good as annotated ones. On difficult photographs, wire extraction limits accuracy; with detected components, missed detections do. Three changes follow from these results: an extractor whose pixel parameters follow the component scale, resampling inputs to a nominal component size, and a detector trained with full rotation augmentation. The VLM reference shows that a general model given component boxes can exceed a geometric join per image on this task. The geometric join remains useful where a local, deterministic and inspectable result is required. A. SCOPE AND LIMITATIONS Task and metric. The pipeline recovers structural connectivity. It does not read component values or device models, so its illustrative netlist export (SPICE syntax) uses placeholder values; simulation of real scans is not evaluated. The component-pair metric ignores terminal identity: a perfect score does not certify pin assignments, the exact net partition or the absence of shorts between pins of one component. The completion guard blocks new same-component merges but does not repair merges made by the base graph. Connectivity

Change 413: replace | old pp. [8] | new pp. [11]

Before (including deletions): by an independent, context-free model call (one per image),

After: from drawn wires only: terminals and supply symbols are wire ends,

Change 414: replace | old pp. [8] | new pp. [11]

Before (including deletions): no answer leaks between circuits,

After: identically labeled supply terminals (e.g., two "+5 V" points)

Change 415: replace | old pp. [8] | new pp. [11]

Before (including deletions): human,

After: implied source between a pair of input terminals are

Change 416: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: interpreted, which would require reading their labels. Table 11 lists

Change 417: replace | old pp. [8] | new pp. [11]

Before (including deletions): model, remains

After: device groups: pin templates for transformers, optocouplers and other complex devices exist but are not validated, and

Change 418: replace | old pp. [8] | new pp. [11]

Before (including deletions): verifier

After: detector merges several original classes into one label. Data. All images come from one corpus. The human- verified nets were bootstrapped from a conservative endpoint- graph join and corrected by a single annotator, which may leave a bias toward graph-based methods; on the held-out benchmark, with labels from the dataset's own annotations, our join again ranks first. The inspection

Change 419: delete | old pp. [8] | new pp. []

Before (including deletions): record. On a clean synthetic control (15 authored circuits)

After: (none)

Change 420: replace | old pp. [8] | new pp. [11]

Before (including deletions): VLM scores F1 = 0.99, confirming it understands

After: 17 overlaps found five human-label errors (four in C242_D1_P1); we left

Change 421: replace | old pp. [8] | new pp. [11]

Before (including deletions): task

After: released human-verified labels unchanged. Only 22 of the 164 held-out images were human-audited,

Change 422: replace | old pp. [8] | new pp. [11]

Before (including deletions): symbol set. On all 31 real scans, evaluated

After: because of the tool's ordering they are mainly smaller circuits; the 17-image validation

Change 423: replace | old pp. [8] | new pp. [11]

Before (including deletions): human-verified net-level ground truth, it reaches micro-F1 = 0.923 (P = 0.97, R = 0.88; macro-F1 = 0.949).

After: human nets (micro-F1 0.939) has more circuits with at least ten electrical components (9 of 17, against 5 of the 22 audited).

Change 424: replace | old pp. [8] | new pp. [11]

Before (including deletions): VLM thus edges our deterministic join on aggregate (0.923 vs 0.890 micro-F1), but

After: reference mainly misses connections, and its quality filter and a junction rule were fixed after inspecting

Change 425: replace | old pp. [8] | new pp. [11]

Before (including deletions): gap is not statistically significant: a paired bootstrap over

After: 17 overlapping images. The benchmark copies of

Change 426: replace | old pp. [8] | new pp. [11]

Before (including deletions): puts the dif- ference at +0.033 with a 95% CI of [-0.009, +0.078],

After: are resized, partly re- oriented, and half stroke maps,

Change 427: replace | old pp. [8] | new pp. [11]

Before (including deletions): includes zero. Two further findings qualify the comparison. First, the VLM's accuracy degrades with circuit complexity: it is exact (F1 = 1.00) on 21

After: makes them easier than raw photographs

Change 428: replace | old pp. [8] | new pp. [11]

Before (including deletions): 31 circuits, including all the small 3–5-component cases, but

After: same drawings. The Hough and CCL baselines keep their 31-image settings

Change 429: replace | old pp. [8] | new pp. [11]

Before (including deletions): largest boards it systematically omits connections, with recall falling

After: held-out photographs. Scale. The join tolerances derive from one median scale per image, and their clamps are tuned

Change 430: replace | old pp. [8] | new pp. [11]

Before (including deletions): 0.63– 0.68 on

After: native resolution. A component much smaller than

Change 431: replace | old pp. [8] | new pp. [11]

Before (including deletions): three densest boards, exactly where

After: median can create

Change 432: replace | old pp. [8] | new pp. [11]

Before (including deletions): netlist error is most costly; its precision otherwise stays high (0.97 micro), so its failure mode is dropped nets rather than shorts. The macro-F1 (0.949) is correspondingly inflated by the many easy small circuits, which is why we report the pair-weighted micro-F1

After: spuri- ous pair,

Change 433: delete | old pp. [8] | new pp. []

Before (including deletions): primary. Second, the cost and form of the output differ from the geometric pipeline: each image consumes on the order of 105 tokens, the result is free-form text rather than a structured netlist, and nothing

After: (none)

Change 434: replace | old pp. [8] | new pp. [11]

Before (including deletions): model guarantees electrical validity (e.g.

After: synthetic Wheatstone case. The extractor's pixel-valued parameters limit accuracy on downscaled inputs. Rescaling tests changes in zoom, not in drawing style, and uses annotated components. Detector. The detector was trained on CGHD images with an 85/15 random split, and no manifest of

Change 435: replace | old pp. [8] | new pp. [11]

Before (including deletions): a two-terminal component is not shorted). Our geometric join is statistically indistinguishable

After: split survives. The 31 benchmark drawings come

Change 436: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: drafters eligible for training, and about 26 of them would be expected in

Change 437: replace | old pp. [8] | new pp. [11]

Before (including deletions): VLM on these circuits while being deterministic, two to three orders of magnitude cheaper, and structurally valid by construction:

After: training split, so

Change 438: replace | old pp. [8] | new pp. [11]

Before (including deletions): self-loop guard guarantees that no two-terminal component is emitted shorted. We therefore treat the VLM as an upper reference

After: end-to-end numbers are likely optimistic

Change 439: replace | old pp. [8] | new pp. [11]

Before (including deletions): what a frontier model can do here, not as a connectivity module to ship: an explicit model reaches the same accuracy and also gives the determinism, low cost, and structural validity a deployed digitizer needs. V. DISCUSSION

After: detection.

Change 440: replace | old pp. [8] | new pp. [11]

Before (including deletions): endpoint-graph model resolves ambiguity that radius- based methods cannot: by representing both wire end- 8

After: detector excludes drafter_0, whose drawing

Change 441: replace | old pp. [9] | new pp. [11, 12]

Before (including deletions): points

After: 11 TABLE 11. Connectivity evaluation, pin construction

Change 442: replace | old pp. [9] | new pp. [12]

Before (including deletions): component

After: export by device group. Device group Evaluated scope Pin construction Export R/C/L/D/Q, voltage sources and ICs Component-pair connectivity on both benchmarks Geometric templates; endpoint clus- tering for discrete and source types, generic layouts for ICs Placeholder values; ICs are skipped because no export model is imple- mented. Switches Scored in the extended held-out vari- ant (Section V-D); not in the primary subset Two

Change 443: replace | old pp. [9] | new pp. [12]

Before (including deletions): as nodes with typed edges,

After: at the shorter OBB-edge midpoints; long-axis fallback A 0.001 Ω resistor if the two pins map to distinct nodes; no switch state is recovered. Transformers, thyristors, optocouplers and other complex devices Not evaluated Two-pin geometric guess for transformers; optocoupler template; generic fallback No export model; skipped. style differs from the rest of the corpus, and

Change 444: replace | old pp. [9] | new pp. [12]

Before (including deletions): distinguishes between a wire that passes near a com- ponent (endpoint–wire-body edge)

After: is sensitive to orientation. Coverage. The human-verified benchmark contains 13 an- notated crossovers, and we did not evaluate dense multi-track buses or multi-layer crossings. The VLM reference uses one model

Change 445: replace | old pp. [9] | new pp. [12]

Before (including deletions): that connects to it (endpoint–pin edge with directional preference). The component-first assignment further disambiguates by using the geometric constraint that wires must reach a component's pin, not just its bounding box. Degree-budget completion addresses the fundamental lim- itation of detector noise: even a 97.6% accurate wire detector drops ~2% of wire endpoints,

After: prompt,

Change 446: replace | old pp. [9] | new pp. [12]

Before (including deletions): those dropped endpoints are precisely the connections that matter most for simulation accuracy. The b-matching formulation bounds the damage from mis-detection (each floating pin gets at most one recov- ery edge), while the self-loop guard prevents the catastrophic short-circuit failures that plague unconstrained approaches. The human-verified net-level evaluation, while novel for hand-drawn circuits, currently covers 31 images; the 95% bootstrap confidence intervals on the per-strategy F1 are cor- respondingly wide (± 0.03 –0.05) and overlap between adja- cent strategies, so while the top ranking is consistent and corroborated by the independently authored synthetic suite, broader annotation is needed to separate the closely ranked configurations. The net-level labels were bootstrapped by a conservative endpoint-graph join over perfect wires and then human-corrected; we mitigate the residual concern that this favors graph-based methods by validating on synthetic ground truth that is authored from scratch, where the same ranking holds. The evaluation draws from a single hand- drawn corpus (CGHD-1152); generalization across other drawing styles, capture conditions, and printed schematics re- mains to be validated, as does extension to circuits containing devices outside the simulatable set (switches, transformers, etc.), which we exclude from the net-level benchmark. The VLM connectivity comparison uses a single frontier model (Claude Opus 4.8) on the same 31 images; evaluating addi- tional models and prompting strategies is future work. Com- ponent detection, at mAP@0.5 = 89.0% and 70.7% recall on crossovers (Section

III-B), also leaves room for improvement; moreover the other class folds 30 rare device types together, so the detector localizes those devices but cannot name them, and a netlist containing one is incomplete. Excluding drafter_0 from training avoids domain-shift artifacts from an outlier drawing style, but leaves that style unmodelled. The pipeline recovers circuit topology, not component values: simulation accuracy is reported

After: was run

Change 447: replace | old pp. [9] | new pp. [12]

Before (including deletions): synthetic suite, where values are known by construction, and no netlist derived from a real scan is simulated here. Producing a numerically simulatable netlist from a scan additionally requires reading component values off the image, which we do not attempt; coupling the join to a value-recognition (OCR) stage, as in [3], [4], and validating end-to-end against a reference simulation, as in [6], is the natural completion of the pipeline. Finally, the

After: human-verified benchmark. The

Change 448: replace | old pp. [9] | new pp. [12]

Before (including deletions): a first-pass approximation of detector noise rather than a

After: authored and not

Change 449: delete | old pp. [9] | new pp. []

Before (including deletions): fit; with the new real-image net-level metric available, calibrating it

After: (none)

Change 450: delete | old pp. [9] | new pp. []

Before (including deletions): the

After: (none)

Change 451: replace | old pp. [9] | new pp. [12]

Before (including deletions): failure distribution is

After: errors. Expansion. The derived reference makes

Change 452: replace | old pp. [9] | new pp. [12]

Before (including deletions): natural next step. VI.

After: larger human-verified benchmark affordable: its nets can seed the verification tool, where checking an image took a median of 77 s in our audit. We plan to sample new images stratified by component count, drafter, crossing and bus structure and capture conditions, have two annotators correct each image independently with adjudication of disagreements, and extend the evaluation to printed schematics and to corpora other than CGHD. VII.

Change 453: replace | old pp. [9] | new pp. [12]

Before (including deletions): present

After: presented

Change 454: delete | old pp. [9] | new pp. []

Before (including deletions): complete

After: (none)

Change 455: insert | old pp. [] | new pp. [12]

Before (including deletions): (none)

After: structural netlist recovery from

Change 456: replace | old pp. [9] | new pp. [12]

Before (including deletions): circuit digitization,

After: schematics,

Change 457: replace | old pp. [9] | new pp. [12]

Before (including deletions): occlusion-first

After: occlusion- first

Change 458: replace | old pp. [9] | new pp. [12]

Before (including deletions): extrac- tor that yields an endpoint representation and

After: extractor,

Change 459: replace | old pp. [9] | new pp. [12]

Before (including deletions): two-stage endpoint-graph join, together with, to our knowledge, the first human-verified net-level connectivity benchmark for hand- drawn circuits. The

After: typed

Change 460: insert | old pp. [] | new pp. [12]

Before (including deletions): (none)

After: and degree- budget completion. With annotated components, the join reaches component-pair micro-F1 0.890 on 31 human- verified images and 0.711 on 164 held-out photographs

Change 461: replace | old pp. [9] | new pp. [12]

Before (including deletions): degree-budget completion is most robust under synthetic error injection (F1 = 0.95

After: independently derived reference nets, and it beats every de- terministic baseline on both with significant paired margins. Its accuracy holds from 0.5× to 3× image scale when wires are given, and it reaches 0.627 end to end with a trained detector. A VLM given the electrical-component boxes scores higher per image,

Change 462: replace | old pp. [9] | new pp. [12]

Before (including deletions): highest severity vs. 0.36 for

After: cost of a proprietary call per image. Better extraction on difficult photographs and better compo- nent detection are

Change 463: replace | old pp. [9] | new pp. [12]

Before (including deletions): radius base- line), and, paired with a high-precision scale-relative base, attains connectivity micro-F1 = 0.890 on real images against human-verified ground truth, beating the prior completion default (0.83) and every deterministic baseline, including the connected-component net-tracing recipe used by recent systems (0.62) and a Hough-plus-proximity line tracer (0.81), all on identical detected wires. Running the join on perfect wires does not raise micro-F1 at all, showing the join, not wire detection, is the bottleneck. A first-party experiment with a frontier VLM reaches 0.923 on the same images, a difference from our join that is not statistically significant (paired 95% CI [-0.009, +0.078]), but at two

After: most direct routes

Change 464: delete | old pp. [9] | new pp. []

Before (including deletions): three orders of magnitude

After: (none)

Change 465: replace | old pp. [9] | new pp. [12]

Before (including deletions): cost and without structurally valid output, supporting the explicit geometric design. Future work includes scaling the human-verified benchmark, calibrating the synthetic error model to the measured detector distribution, and extending coverage to devices outside the simulatable set.

After: end-to-end accuracy.

Change 466: replace | old pp. [9] | new pp. [12]

Before (including deletions): IV-F,

After: V-I,

Change 467: replace | old pp. [9] | new pp. [12]

Before (including deletions): evaluation scripts,

After: the

Change 468: replace | old pp. [9] | new pp. [12]

Before (including deletions): genera- tor,

After: generator,

Change 469: replace | old pp. [9] | new pp. [12]

Before (including deletions): (ground_truth/real_nets_verified.json),

After: (ground_truth/real_net

Change 470: insert | old pp. [] | new pp. [12]

Before (including deletions): (none)

After: CGHD-derived reference nets of the held-out benchmark (ground_truth/cghd_ref/), the human-audited nets of 22 held-out images with the verification timing log (ground_truth/cghd_ref_audit/real_nets_working. and timing_log.jsonl), the

Change 471: replace | old pp. [9] | new pp. [12]

Before (including deletions): connected-component baseline

After: base- line

Change 472: delete | old pp. [9] | new pp. []

Before (including deletions): and

After: (none)

Change 473: replace | old pp. [9] | new pp. [12]

Before (including deletions): harness

After: har- ness and the evaluation scripts for the paired tests, strata, rescaling, held-out and end-to-end experiments (wire_detection/benchmark/revision2/)

Change 474: replace | old pp. [9] | new pp. [12]

Before (including deletions): <https://github.com/boscochanam/circuit-digitization>; an archived snapshot of

After: <https://github.com/boscochanam/circuit-digitization>, together with

Change 475: replace | old pp. [9] | new pp. [12]

Before (including deletions): v1.0.1 release is deposited at Zenodo (doi: 10.5281/zenodo.21274159).

After: image-to-drafter mapping used in Ta- ble 10. The reference nets are derived from the CGHD annotations [14]; no CGHD images are redistributed.

Change 476: replace | old pp. [9] | new pp. [12]

Before (including deletions): <https://huggingface.co/boscochanam/circuit-component-detector>,

After: <https://huggingface.co/boscochanam/circuit-component-detector>,

Change 477: replace | old pp. [9] | new pp. [12]

Before (including deletions): the CGHD-1152 dataset [1], [2]. The evaluation

After: CGHD

Change 478: replace | old pp. [9] | new pp. [12]

Before (including deletions): are drawn from CGHD-1152, which is distributed under a CC BY 4.0 licence [2]; our released overlays are adaptations of those images (grayscale conversion, upscaling, and ground-truth wire polylines drawn on top) and are redistributed under the same licence, while our net annotations are original work released under the MIT licence. VOLUME 11, 2023 9

After: [14].

Change 479: delete | old pp. [10] | new pp. []

Before (including deletions): F. Thoma, J. Bayer, Y. Li, and A. Dengel, "A public ground-truth dataset for handwritten circuit diagram images," in Proc. Int. Conf. Document Analysis and Recognition (ICDAR), 2021, pp. 20–27. [Online]. Available: https://doi.org/10.1007/978-3-030-86198-8_2 [2] F. Thoma, J. Bayer, and Y. Li, "CGHD1152," Zenodo, 2022, dataset, licensed CC BY 4.0. [Online]. Available: <https://doi.org/10.5281/zenodo.6385814> [3] C. R. Kelly and J. M. Cole, "Digitizing images of electrical-circuit schematics," APL Mach. Learn., vol. 2, no. 1, p. 016109, 2024. [4]

After: (none)

Change 480: replace | old pp. [10] | new pp. [12]

Before (including deletions): [5] J. Bhandari, V. Bhat, Y. He, H. Rahmani, S. Garg, and R. Karri, "Masala- CHAI: A large-scale SPICE netlist dataset for analog circuits by harnessing AI," arXiv:2411.14299, 2024. [6]

After: [2]

Change 481: replace | old pp. [10] | new pp. [12]

Before (including deletions): [7]

After: [3] J. Bhandari, V. Bhat, Y. He, H. Rahmani, S. Garg, and R. Karri, "Masala- CHAI: A large-scale SPICE netlist dataset for analog circuits by harnessing AI," arXiv:2411.14299, 2024. [4] C. R. Kelly and J. M. Cole, "Digitizing images of electrical-circuit schematics," APL Mach. Learn., vol. 2, no. 1, p. 016109, 2024. [5]

Change 482: replace | old pp. [10] | new pp. [12]

Before (including deletions): [8]

After: [6]

Change 483: replace | old pp. [10] | new pp. [12]

Before (including deletions): [9]

After: [7]

Change 484: replace | old pp. [10] | new pp. [12]

Before (including deletions): [10]

After: [8]

Change 485: replace | old pp. [10] | new pp. [12]

Before (including deletions): [11]

After: [9]

Change 486: replace | old pp. [10] | new pp. [12, 13]

Before (including deletions): [12]

After: 12 VOLUME 11, 2023 [10]

Change 487: replace | old pp. [10] | new pp. [13]

Before (including deletions): [13]

After: [11]

Change 488: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: [12] S. Gao, F. Shi, X. Xu, J. Chen, H. H.-C. lu, et al., "A memristor- based neural network circuit with classical conditioning and fear gen- eralization," IEEE Trans. Consumer Electron., early access, 2026, doi: 10.1109/TCE.2026.3663359.
[13] S. Gao, Y. Song, Y. Cao, H. H.-C. lu, L. Qin, Y. Zhang, and J. Mou, "Biologically plausible memristive decision-making circuit for adaptive control in industrial autonomous navigation," IEEE Trans. Ind. Informat., vol. 22, pp. 4801–4812, 2026, doi: 10.1109/TII.2026.3668794.

Change 489: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: F. Thoma, J. Bayer, Y. Li, and A. Dengel, "A public ground-truth dataset for handwritten circuit diagram images," in Proc. Int. Conf. Document Analysis and Recognition (ICDAR), 2021, pp. 20–27. [Online]. Available: https://doi.org/10.1007/978-3-030-86198-8_2 [15]

Change 490: replace | old pp. [10] | new pp. [13]

Before (including deletions): [15]

After: [16]

Change 491: replace | old pp. [10] | new pp. [13]

Before (including deletions): [16]

After: [17]

Change 492: replace | old pp. [10] | new pp. [13]

Before (including deletions): [17]

After: [18]

Change 493: replace | old pp. [10] | new pp. [13]

Before (including deletions): [18]

After: [19]

Change 494: replace | old pp. [10] | new pp. [13]

Before (including deletions): Kulkarni

After: Kulkarni, V. Kumar, R. L. Minz, M. Varshney, T. Samon, A. Mitra,

Change 495: replace | old pp. [10] | new pp. [13]

Before (including deletions): ICCV Workshops,

After: IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW),

Change 496: replace | old pp. [10] | new pp. [13]

Before (including deletions): [19]

After: [20]

Change 497: replace | old pp. [10] | new pp. [13]

Before (including deletions): [20]

After: [21]

Change 498: replace | old pp. [10] | new pp. [13]

Before (including deletions): [21]

After: [22]

Change 499: replace | old pp. [10] | new pp. [13]

Before (including deletions): [22]

After: [23]

Change 500: replace | old pp. [10] | new pp. [13]

Before (including deletions): [23]

After: [24]

Change 501: replace | old pp. [10] | new pp. [13]

Before (including deletions): [24]

After: [25]

Change 502: replace | old pp. [10] | new pp. [13]

Before (including deletions): [25]

After: [26]

Change 503: replace | old pp. [10] | new pp. [13]

Before (including deletions): [26]

After: [27]

Change 504: replace | old pp. [10] | new pp. [13]

Before (including deletions): [27]

After: [28]

Change 505: delete | old pp. [10] | new pp. []

Before (including deletions): 10 VOLUME 11, 2023

After: (none)

Change 506: insert | old pp. [] | new pp. [13]

Before (including deletions): (none)

After: VOLUME 11, 2023 13

Change 507: insert | old pp. [] | new pp. [14]

Before (including deletions): (none)

After: 14

Change 508: delete | old pp. [11] | new pp. []

Before (including deletions): 11

After: (none)